Supplemental Information for

Comparative genomics provides new insights into the aquatic adaptations of mammals

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Materials and Methods

Sample collection, sequencing library preparation and sequencing

DNA of 15 species [*Kogia sima*, *Kogia breviceps*, *Pseudorca crassidens*, *Lagenorhynchus obliquidens*, *Grampus griseus*, *Tursiops aduncus*, *Zalophus californianus*, *Otaria byronia*, *Arctocephalus australis*, *Phoca vitulina*, *Phoca largha*, *Pusa sibirica*, *Balaenoptera brydei*, *Mesoplodon densirostris* and *Neophocaena phocaenoides*, *Balaenoptera brydei*, *Mesoplodon densirostris*, and *Neophocaena phocaenoide*] were extracted using cetyl trimethylammonium bromide (CTAB)(1). A skin sample of *Lagenorhynchus australis* was collected using a Paxarm biopsy system (NEW ZEALAND LTD) in Golfo San Rafael, Chile and extracted using phenol chloroform (2). DNA quality control was performed by using Pulsed-Field Gel Electrophoresis Systems (CHEF MAPPER，Bio-Rad) with the parameters 'Run time: 16 hours, Cooling temperature: 14 ℃, Volts/CM: 6V/CM, Initial switch time: 0.22s, Final switch time: 44.69s'.

For the blue whale (*B. musculus*), a skin biopsy sample was obtained from a free-ranging animal using a Larsen gun (Palsbøll et al. 1991). The sample was obtained in Skjálfandi Bay off the coast of the town Húsavík in Iceland at GPS coordinates: 66.18517, -017.51138 in 2015. It was imported to Denmark for processing under Article VII; paragraph 6 CITES convention for import as scientific exchange between CITES institution Natural History Museum of Denmark (DK-003) and University of Iceland (IS 005). Total genomic DNA of the blue whale was extracted from the skin biopsy (two replicates), avoiding the white subcutaneous layer, using the DNeasy Blood and Tissue kit (QIAGEN, Hilden, Germany), with the following modifications to increase DNA yields: samples were eluted in 50 µl AE and incubated for 10 minutes at 37 °C. DNA concentration and integrity were verified using an Agilent Genomic DNA ScreenTape (Waldbronn, Germany).

Sample collection and use protocols were approved by the Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, with the ethical statement number IDSSE-SYLL-MMMBL-01.

WGS libraries

WGS libraries with an average insert size of 350 bp were generated for 12 species [*Kogia sima*, *Kogia breviceps*, *Pseudorca crassidens*, *Lagenorhynchus obliquidens*, *Grampus griseus*, *Tursiops aduncus*, *Zalophus californianus*, *Otaria byronia*, *Arctocephalus australis*, *Phoca vitulina*, *Phoca largha*, and *Pusa sibirica*] using a MGI Easy FS PCR-Free DNA Library Prep Set kit (MGI, China). Paired-end reads (100 bp) were obtained on the BGISEQ-500 platform. Paired-end short insert libraries and mate-paired libraries for *Balaenoptera brydei*, *Mesoplodon densirostris*, and *Neophocaena phocaenoides* were sequenced on the Illumina HiSeq 2,000 platform (150 bp reads).

10X Genomics Chromium Genome library preparation

For preparation of the Chromium library, 1 ng of high-quality DNA was denatured according to standard protocol of Genome Library Kit & Gel Bead Kit v2 (part no. 120258). After PCR, the standard circularization step for BGISEQ-500 was carried out, and DNA nanoballs (DNBs) were prepared (3). Paired-end reads with a length of 150 bp were generated on the BGISEQ-500 platform.

Hi-C library preparation

For muscle samples, tissues were cut into ~0.5 cm × 0.5 cm × 0.5 cm cubes and ground into fine powder using liquid nitrogen. Next, samples were fixed in 1% formaldehyde solution, and glycine solution was added used to stop the reaction. For blood samples, fresh blood cells (~5×10^6) were collected, centrifuged, and resuspended in 1 ml 1× PBS solution. Cells were then cross-linked by adding formaldehyde, followed by the addition of 2.5M glycine solution to stop the reaction.

Muscle and blood samples were next processed using the same pipeline. The formaldehyde-fixed samples were suspended in nuclei isolation buffer (10 mM Tris-HCl pH 8.0 (Sigma; St. Louis, MO), 10 mM NaCl (Beyotime, Shanghai, China), 1 × PMSF (Sigma), and incubated in 0.5% SDS at 62 °C for 10 mins. Next, 10% Triton X-100 (Sigma) was added, and each sample was centrifuged to collect nuclei. DNA was digested by restriction enzyme (*Mbo I*) (NEB, Ipswich, MA), and the 5’ overhang was repaired using a biotinylated residue (0.4 mM biotin-14-dATP (Invitrogen, Carlsbad, CA). The resulting blunt-ended fragments were ligated *in*

situ (10× T4 DNA ligase buffer (NEB), 10% Triton X-100 (Sigma), 10 mg/ml BSA (NEB), T4 DNA ligase (NEB). Finally, the isolated DNA was reverse-crosslinked [by adding 10 mg/ml proteinase K (NEB) and 1% SDS (Invitrogen) to the tube followed by incubation at 56℃ overnight]. The reverse-crosslinked DNA was aliquoted into three tubes and purified as follows: addition of 1.5 × volumes of AMpure XP (Agencourt Bioscience, Beverly, MA) to each tube to precipitate onto beads [vortexing and spinning down briefly, incubating for 10 min. at room temperature], harvesting of beads with a magnetic particle separator (MPS) (Invitrogen) for 5 min. at room temperature, discarding supernatant, washing the beads twice with 1 ml of freshly made 75% ethanol (Sinopharm, Shanghai, China), air-drying the beads completely, and resuspending the beads in 30 µl of double-distilled water (ddH_2O). The Hi-C library was created by shearing 20 µg of DNA and capturing the biotin-containing fragments on streptavidin-coated beads using Dynabeads MyOne Streptavidin T1 (Invitrogen). The DNA fragments were end-repaired, and adaptor ligation was performed. After PCR [95℃ 3 min, eight cycles of 98℃ 20 sec., 60℃ 15 sec., and 72℃ 15 sec] with a final extension at 72℃ for 10 min, the Hi-C libraries were sequenced on the BGISEQ-500 platform.

ATAC library preparation

This protocol is highly similar to a previously published protocol (4), and either protocol can be used for the isolation of nuclei with equivalent results.

1. Isolation of nuclei from frozen samples for ATAC-seq

   Place frozen samples (500 mg muscle or 200 µl blood) into a pre-chilled 2 ml Dounce homogenizer with 1.5 ml cold 1× Homogenization Buffer [320 mM sucrose, 0.1 mM EDTA, 0.1% NP40, 5 mM $CaCl_2$, 3 mM $Mg(Ac)_2$, 10 mM Tris pH 7.8, 1× protease inhibitors (Roche cOmplete], and 167 µM -mercaptoethanol; in water]. Allow frozen tissue to thaw for 10 minutes until the tissue chunk sinks to the bottom of the Dounce (i.e., mortar). Homogenize until ‘resistance’ goes away (~10 strokes). Muscle tissue must be filtered using a 100 µm nylon mesh before further homogenization. Homogenize for a further 20 strokes. Transfer 400 µl to a round-bottom 2 ml Lo-Bind Eppendorf tube. Add 1 volume (400 µl) of 50% Iodixanol solution (50% iodixanol in 1× homogenization buffer), to give a final concentration of

25% iodixanol, and mix by pipetting. Layer 600 µl of 29% iodixanol solution (29% iodixanol in 1× homogenization buffer containing 480 mM sucrose) under the 25% mixture. Avoid mixing layers. Layer 600 µl of 35% iodixanol solution (35% iodixanol in 1× homogenization containing 480 mM sucrose) under the 29% mixture. Avoid mixing layers. In a swinging bucket centrifuge, spin for 30 min at 3,000 RCF. Aspirate the top layers down to within 300 µl (the nuclei band). Using upper 200 µl volume, collect the nuclei band and transfer to a fresh tube. Count nuclei using DAPI staining and aliquot nuclei for ATAC reaction.

2. Transposition reaction

Transfer 2 $\times 10^5$ nuclei into a tube containing 1 ml of ATAC-RSB+0.1% Tween-20. Centrifuge nuclei for 10 minutes at 500g at 4℃ for twice. Aspirate supernatant and add reaction mix (10 µl 5× TAD buffer, 2.5 µl transposase, 36.5 µl PBS, 0.5 µl 1% digitonin, 0.5 µl 10% Tween-20,5 µl H_2O) to pellet. Resuspend by pipetting up and down 10 times. Incubate reaction at 37ºC for 30 minutes in a thermomixer with 1000 RPM mixing. Cleanup reaction with DNA MinElute Kit (QIAGEN). Elute DNA in 20 µl elution buffer and store at -20°C until ready to amplify.

3. Preamplification of transposed fragments

Amplify for 8 cycles using NEBNext 2× Master Mix (NEB). Each reaction should contain 2.5 µl of 25 µM N5 primer, 2.5 µl of 25 uM N7 barcode primer, 25 µl 2× NEBNext master mix, and 20 µl transposed/cleaned up sample. Thermocycle (72ºC for 5 min, 98ºC for 30 sec, followed by 8 cycles of [98ºC for 10 sec, 63ºC for 30 sec, 72ºC for 1 min] then hold at 4ºC. After amplification is complete, remove tubes from the thermocycler and at 4℃.

4. qPCR amplification to determine additional cycles:

Using 5 µl (10%) of the pre-amplified mixture, run a 20 µl qPCR to determine the number of additional cycles needed. The conditions for this 20 µl volume PCR are 4.12 µl nuclease-free water, 0.4 µl 20 µM Tn5 AD153F primer, 0.4 µl 25 uM Tn5 AD153R primer, 0.08 µl ROX Reference Dye (50×), 10 µl SYBR-Premix Ex TaqII (2×), 5 µl pre-amplified sample. Cycling conditions for this PCR are 98ºC for 30 sec, followed by 40 cycles of [98ºC for 10 sec, 63ºCfor 30 sec, 72ºC for 1 min]. After qPCR amplification, manually assess the amplification profiles and determine the required number

of additional cycles. Commonly, to calculate the number of additional cycles needed, plot linear Rn versus cycle and determine the cycle number that corresponds to one-third of the maximum fluorescence intensity.

5. Final amplification and cleanup:

   Using the remainder of the pre-amplified DNA (~45ul), run the required number of additional cycles. Place the pre-amplified tubes back into the thermocycler. Purify the final PCR reaction using a DNA MinElute Kit (QIAGEN) and elute in 20 µl ddH_2O. Purify libraries using 0.5-0.7× AMPure XP Beads (Beckman Coulter) size selection (~150-700 bp), then 4 libraries pooled together.

6. Standard circularization step:

   Input 330ng pooled DNA for sequencing on the BGISEQ-500 platform.

Genome assembly

Raw reads were filtered [using SOAPnuke (v1.5.6) (5) with parameters 'filter -l 10 -q 0.2 -d -n 0.05 -Q 2 --misMatch 1 --matchRatio 0.2'] to remove reads with an ambiguous base content higher than 5%, low-quality bases (quality score less than 10) higher than 20%. For raw data generated using Hi-C libraries, the HiC-Pro pipeline (*see below*) was used.

For genomes generated using HiSeq, Platanus (v1.2.4) (6) was used to assemble reads using default parameters. After conversion to a compatible format, clean 10X Genomics reads were *de novo* assembled using Supernova (v1.2.0) (7) with default parameters. The 'pseudohap' style was used to output the Supernova final assembly (scaffolds with length >100 bp were retained). Next, ARKS (8) was used to link scaffolds. Gaps were filled using KGF (v1.19) and GapCloser (v1.10) (9) with default parameters. Raw FASTQ files from Hi-C derived libraries were quality controlled using the HiC-Pro (v2.8.0_devel) (10) pipeline with default parameters. Finally, 10X Genomics assemblies were anchored to chromosomes using the 3D-DNA pipeline (v170123) (11) and the parameters '-m haploid -s 4' with valid read pairs.

Next, we carried out *k*-mer analysis with ~50× clean reads from each species. The estimated genome size (*G*) was calculated by using the formula G = num of kmer/peak-depth.

Genome annotation

Annotation of repeat elements was performed by combining homology-based and *de novo* prediction strategies. Firstly, we identified repeats against the Repbase (12) database by using RepeatMasker (v4.0.5) (13) [parameters '-norna -engine ncbi -nolow -no_is'] and RepeatProteinMasker (v4.0.5) [parameters '-engine ncbi -noLowSimple -pvalue 0.0001'] at the nucleic acid and protein level, respectively. Secondly, *de novo* prediction was carried out using RepeatModeler (v1.0.8) and LTR-FINDER (v1.0.6) (14) with default parameters. Thirdly, Tandem Repeat Finder (v4.07) [parameters '-Match 2 -Mismatch 7 -Delta 7 -PM 80 -PI 10 -Minscore 50 -MaxPeriod 2000'] was used to detect tandem repeats.

Before gene annotation, assembled genomes were repeat masked. Gene annotation was also performed by combining homology-based and *de novo* prediction methods. Firstly, AUGUSTUS (v3.1) (15) was utilized, with the human as the training set, for *de novo* annotation. As for homolog-based prediction, we downloaded protein sequences of bottlenose dolphin (*Tursiops truncatus*), Yangtze finless porpoise (*Neophocaena asiaeorientalis*), minke whale (*Balaenoptera acutorostrata*), sperm whale (*Physeter microcephalus*), baiji (*Lipotes vexillifer*), killer whale (*Orcinus orca*), northern fur seal (*Callorhinus ursinus*), Antarctic fur seal (*Arctocephalus gazelle*), Hawaiian monk seal (*Neomonachus schauinslandi*), sheep (*Ovis aries*), cattle (*Bos taurus*), dog (*Canis lupus familiaris*), and human (*Homo sapiens*) from NCBI Annotation Release 105 (16). Candidate gene loci were firstly identified by aligning these protein sequences to assembled genomes using BLAT (17). Gene models were then further predicted using GeneWise (v2.4.1) (18) with default parameters. Finally, we integrated all the homolog predicted genes and *de novo* detected genes using GLEAN (19) software with default parameters to obtain non-redundant gene sets. We assessed these gene sets using BUSCO (v2.0) (20) and its mammalia_odb9 lineage dataset. To assess the function of the gene sets, we mapped them to Kyoto Encyclopedia of Genes and Genome (KEGG v84.0) (21), Swiss-Prot (v2017_09) (22), and TrEMBL (v2017_09) (23) databases using BLASTp (v 2.2.26) (24) and an *E*-value cutoff of 1×10^{-5}. Gene domains and motifs were predicted using InterProScan (v5.16-55.0) (25) against ProDom (26), Pfam (27), SMART (28), PANTHER (29), and PROSITE (30). Gene ontology (GO) (31) terms were extracted from the InterProScan results.

Phylogenetic tree construction

To clarify the phylogenetic relationships of three marine mammal lineages, we generated two datasets from different resources were used to infer the phylogenetic tree. Firstly, whole-genome alignments were created from 35 marine mammal genomes (Cetacea: 23; Pinnipedia: 11; Sirenia: 1) and 16 outgroup terrestrial mammals with the LAST (version 885) (32) and MULTIZ (version 11.2) (33), and we resample a total of 2771 synteny alignment chunks, and ensured that there is at least 20 Kb between each two chunks. Secondly, we applied the reciprocal best hit (RBH) method to identify the genes that are 1:1 orthologous among all the 51 species and got a total of 596 ortholog pairs. For each ortholog pair, we extracted the longest exon to represent the gene to avoid the mixing of multiple c-genes. These extracted sequences are also having a mean distance of 12 Mb and a minimum of 20 Kb from each other, ensuring that they are independent and single c-genes. In order to ensure the reliability of each gene tree, we have made the following efforts: 1) To avoid model misspecification, we applied IQ-TREE (34) for the inference of each alignment, which could automatically select the suitable model for tree inference, we also employ aLRT and bootstrap test to collapse branches on each tree inference process. 2) For each dataset (the synteny alignment dataset and the RBH protein dataset), we have tried both the strict strategy (require the c-gene with a minimum bootstrap threshold of 75% for every node) and the loose strategy (included all the c-genes) for subsequent summary coalescence analysis. All the results support a same topology. 3) For each dataset (the synteny alignment dataset and the RBH protein dataset), we calculated the Robinson-Foulds (RF) distances between the topology from each alignment and the summarized topology. We then manually checked the top 10 alignments with the highest RF distances from each dataset. It is confirmed that they are the correct homolog sequences across the 52 species and the highest RF distances were added in the supplementary material **(Figure S6)**.

Species tree inference

We reconstructed the species tree using two data sets: the synteny alignment dataset, and the RBH protein dataset. For each chunk/gene, phylogenetic trees were computed using IQ-TREE with the following parameters: “-nt 3 -st AA -m MFP

-bb 1000 –bnni". We constructed the species tree from the collection of the ML gene trees using ASTRAL-III (35) with a gene-only bootstrapping procedure under the MSC model.

Divergence time calibration

Divergence times were estimated using MCMCtree in PAML v4.8 (36) with the topology of the 4DTV positions and five calibration time points based on the TimeTree database (37): opossum vs. platypus (163.0~191.0 Mya), hippopotamus vs. blue whale (52.4~66.0 Mya), sperm whale vs. blue whale (36.4.6~52.4 Mya), Blainville's beaked whale vs. Yangtze river dolphin (13.2~23.0) polar bear vs. sea otter (37.0~43.0 Mya), and California sea lion vs. walrus (16.8~22.1 Mya).

Demographic history reconstruction

We inferred the demographic history of 31 marine mammals by applying the Pairwise Sequentially Markovian coalescent (PSMC) method (38). To ensure the quality of consensus sequences, we only used samples with more than 30× sequencing depth for each species. Consensus sequences were obtained using SAMtools and the parameters 'mpileup -q 20 -Q 20' and divided into non-overlapping 100 bp bins. PSMC analyses were performed using the following parameters: -N25 -t15 -r5 -p "4+25*2+4+6". The estimated generation time (g) of some marine mammals was obtained from the International Union for Conservation of Nature (IUCN) Red List (https://www.iucn.org). The neutral mutation rate (μ) was estimated using r8s **(Table S10)** (39). To explore effective population size (*Ne*) dynamics across multiple species, we calculated the normalized *Ne* by dividing the real value of *Ne* at each time point with the maximum value for the same species throughout the time scale.

Construction of ancestor chromosome karyotype

For the reconstruction of ancestral chromosomes for marine mammals, cetaceans, and Odontoceti, we selected IPB dolphin as a reference genome and the human genome as outgroup. For ancestral chromosomes of Cetacea and Ruminantia, we selected cattle as the reference genome and human as outgroup. To reconstruct the ancestral chromosomes for Pinnipedia, Carnivora, and Otariidae, we

selected South American sea lion as the reference genome and human as outgroup. To reconstruct ancestral chromosomes for Phocidae, we selected the Baikal seal as the reference genome and human as outgroup. We generated our pairwise alignments using LASTZ (40) (version 1.04.00) with the parameters 'T=2 C=2 H=2000 Y=3400 L=6000 K=2200'. Next, the pairwise alignment results were converted into the UCSC "chain" and "net" formats using axtChain, chainSort, chainPreNet, chainNet, and netSyntenic (http://hgdownload.cse.ucsc.edu/admin/exe/) with default parameters. DESCHRAMBLER (41) was used for constructing chromosomes at 300 kb resolution.

Identification of marine mammal ultra-conserved elements

We used the genomes of 19 marine mammals to identify conserved non-coding elements (CNEs) in cetaceans and pinnipeds. Terrestrial mammal genomes of 29 species (*Ursus maritimus* (GCF_000687225.1), *Enhydra lutris* (GCF_002288905.1), *Homo sapiens* (GCF_000001405.38), *Mus musculus* (GCF_000001635.26), *Bos taurus* (GCF_000003055.6), *Ovis aries* (GCF_000298735.2), *Equus caballus*(GCF_000002305.2), *Sus scrofa* (GCF_000003025.6), *Canis lupus familiaris* (GCF_000002285.3), *Loxodonta africana* (GCF_000001905.1), *Macaca mulatta* (GCF_000772875.2), *Felis catus* (GCF_000181335.3), *Acinonyx jubatus* (GCF_003709585.1), *Ailuropoda melanoleuca* (GCF_000004335.2), *Panthera tigris altaica* (GCF_000464555.1), *Oryctolagus cuniculus* (GCF_000003625.3), *Capra hircus* (GCF_001704415.1), *Monodelphis domestica* (GCF_000002295.2), *Erinaceus europaeus* (GCF_000296755.1), *Heterocephalus glaber* (GCF_000247695.1), *Sorex Araneus* (GCF_000181275.1), *Galeopterus variegatus* (GCF_000696425.1), *Dasypus novemcinctus* (GCF_000208655.1), *Manis javanica* (GCF_001685135.1), *Urocitellus parryii* (GCF_003426925.1), *Orycteropus afer* (GCF_000298275.1), *Desmodus rotundus* (GCF_002940915.1), *Vulpes vulpes* (GCF_003160815.1) and *Mustela putorius furo* (GCF_000239315.1)) were downloaded from NCBI database were used as outgroups.

To identify the CNEs of each group, we initially used LASTZ (40) and chainNet to generate pairwise alignments. For cetacean genomes, we used *Tursiops aduncus* as the reference; and for pinnipeds, we used *Pusa sibirica*. We then used MULTIZ

(http://www.bx.psu.edu/miller_lab/) to generate multiple sequence alignments. We detected sequences with 100% identity (without insertion and deletion) among cetaceans and pinnipeds with a sequence length of at least 20 bases. These CNEs were then aligned to outgroup genomes using BOWTIE2 (http://bowtie-bio.sourceforge.net/index.shtml) (42) with default parameters. Aligned sequences were removed, together with the duplicated sequences, to obtain cetacean- and pinniped- specific CNEs. We used Bowtie2 to align CNEs to annotated genomes, retrieving mapping coordinates with respect to genes: 5' 3kb (relative to the translation start site); 3' 3kb (relative to the translation stop site), coding region; intronic; and intergenic.

Expansion and contraction of gene families

To examine the expansion and contraction of gene families in marine mammals, protein sequences were obtained from three distinct groups of marine mammals [the cetaceans blue whale, sperm whale, Indo-Pacific bottlenose dolphin; the sirenian Florida manatee; the pinnipeds Pacific walrus, Baikal seal, California sea lion] and terrestrial mammals (human, goat, dog, African elephant, opossum), which were then clustered into gene families using OrthoFinder (v1.0.6) (43). A total of 259,215 protein sequences were clustered into 22,239 gene families.

Gene family expansion or contraction analyses were performed using CAFÉ (44). For three branches and nodes of marine mammal, an "expanded and contracted gene family" with both an overall *P*-value (family-wide *P*-value) and an exact *P*-value ≤ 0.01 was defined as a "significantly expanded and contracted gene family.

PSGs and REGs in three groups of marine mammals

We used a conserved genome synteny methodology to establish a high-confidence orthologous gene set that includes cattle, dog, African elephant, American mink, greater horseshoe bat, human, hippopotamus, horse, opossum, pig, polar bear, sea otter, and 35 marine mammal species. Briefly, pairwise whole-genome alignments were constructed using LAST, with the goat genome serving as the reference genome. To minimize the effect of errors [attributed to annotation, sequencing, and assembly], pseudogenes, non-orthologous alignments, and non-conserved gene structures on subsequent evolutionary rate analyses, a series of rigorous filtering criteria were adopted: (1) genes must map to goat

genome via a single chain of sequence alignments, including at least 80% of its coding sequence (CDS), and met the alignment length/score thresholds required for inclusion in the MULTIZ alignments; (2) frame-shift indels in CDSs were prohibited; (3) CDS with premature stop codons were excluded; (4) CDS less than 150 bp of informative site in all species were excluded.

To find genes under positive selection, we applied three different branch-site likelihood ratio tests: for the branch leading to cetaceans, the branch leading to pinnipeds, and the manatee lineage. In all tests, the ratio of non-synonymous to synonymous substitutions (d_N/d_S) for each branch was estimated using the free ratio model of Codeml in PAML v4.8 (36) using default parameters and the branch-site likelihood ratio test to identify positively selected genes in the cetacean branch, the pinniped branch, and the manatee lineage (a sirenian) respectively. An LRT (likelihood ratio test) was performed to compare a model that allows sites to be under positive selection on the foreground branch with the null model in which sites may evolve either neutrally and under purifying selection. *P*-values were computed based on the χ^2 statistics, and genes with *P*-values less than 0.05 were treated as candidates of positive selection.

We applied the branch model in PAML v4.8 (36) with the null model and the alternative model to detect rapidly evolving genes (REGs). Genes with a *P*-value less than 0.05 and a higher ω value in the foreground branches were considered REGs.

Identification of marine mammal specific amino acid mutations and convergence signals.

The multiple sequence alignments of orthologous proteins of marine mammals and terrestrial mammals allowed us to search for convergent amino acid residues. Due to the variation of quality among the different genomes, there were some missing sites in the protein sequences of the orthologous genes. The ancestral sequences for orthologous genes with at least 75% valid amino acid information were reconstructed with the Codeml program in PAML v4.8 (36). We applied a strict *convergence at conservative sites* (CCS, for short) method to detect convergence. Compared with outgroups, convergence is inferred at conservative sites where more than 90% target species shared the same amino acid substitution.

Evolution of specific traits in cetaceans

Giant size

In order to identify candidate genes contributing to giant size in cetaceans, we selected 1,528 genes from the KEGG pathway database annotated with the development of body size. We assigned blue whale and sperm whale to a large size whale group (LSWG) and assigned elephant, human, horse, pig, cattle, dwarf sperm whale, Pygmy sperm whale and Indo-Pacific bottlenose dolphin to the background group.

Extreme diving

We collected a total of 2,289 genes from HypoxiaDB (http://www.hypoxiadb.com), a database of hypoxia-regulated proteins (45). Of these genes, 1,803 was present in our orthologous gene dataset. Sperm whale, dwarf sperm whale, pygmy sperm whale, and Blainville's beaked whale were assigned into the deep-diving group, while twenty shallow diving marine mammal species were assigned to the background group.

Echolocation

We collected 597 genes related to sound in the GO annotation database, of which 504 could be retrieved from our orthologous gene dataset. We compared 40 species – including five echolocating bats (Big brown bat, Natal long-fingered bat, Brandt's bat, Little brown bat, Greater horseshoe bat), and sixteen echolocating cetaceans (toothed whales). Different type of groups are used to detect convergence for comparison.

Supplementary Figs

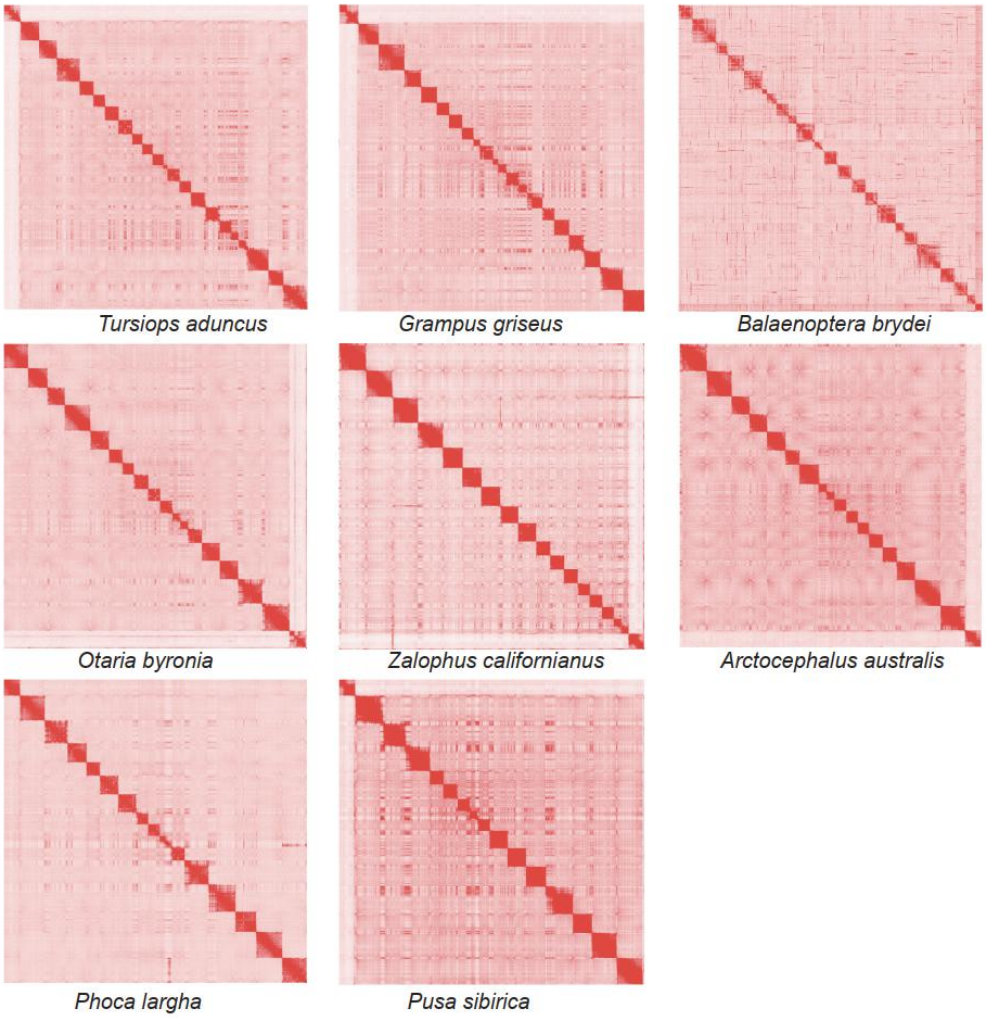


Fig. S1. Hi-C contact maps of eight marine mammal species plotted at 125 kb resolution.

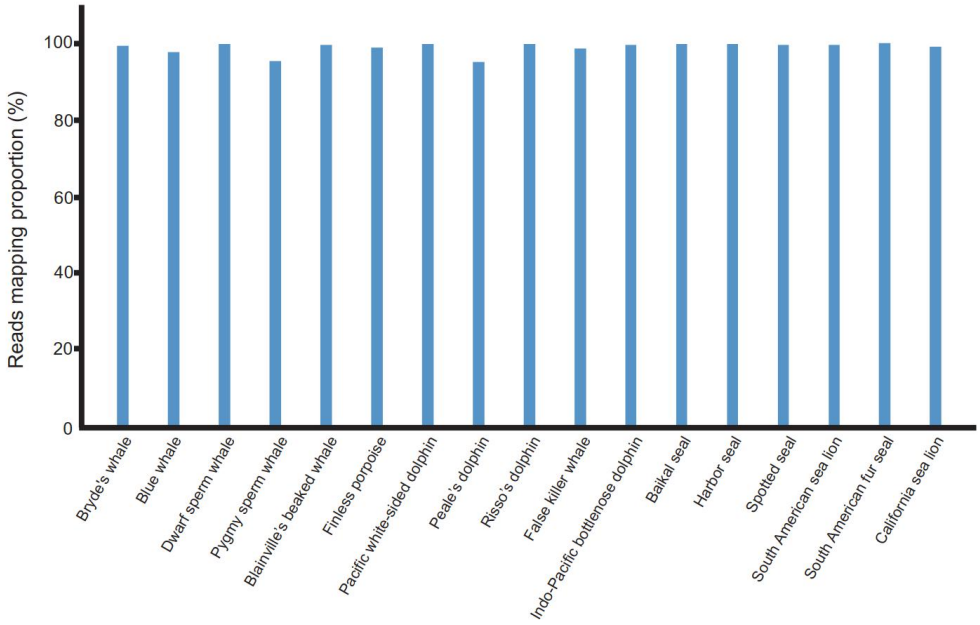


Fig. S2. The proportion of short reads mapped to respective genome assembly for 17 marine mammals.

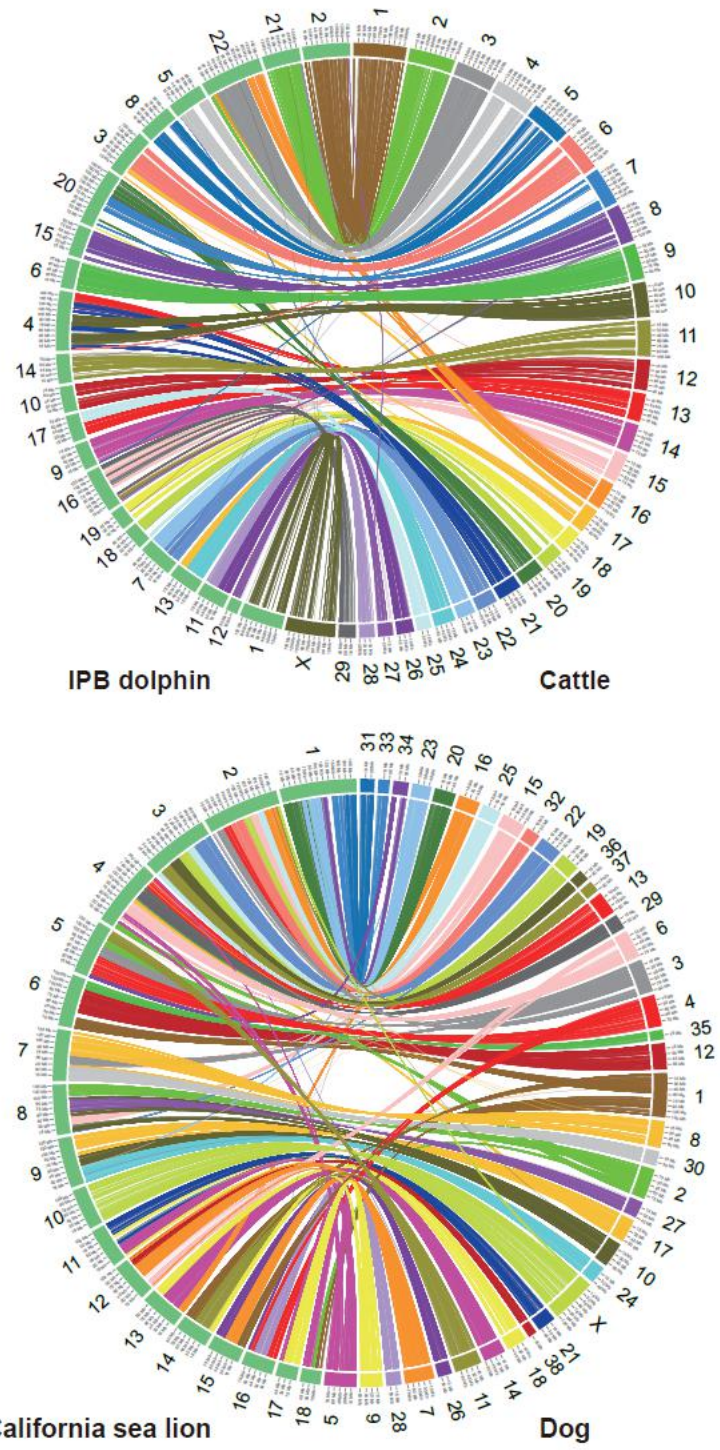
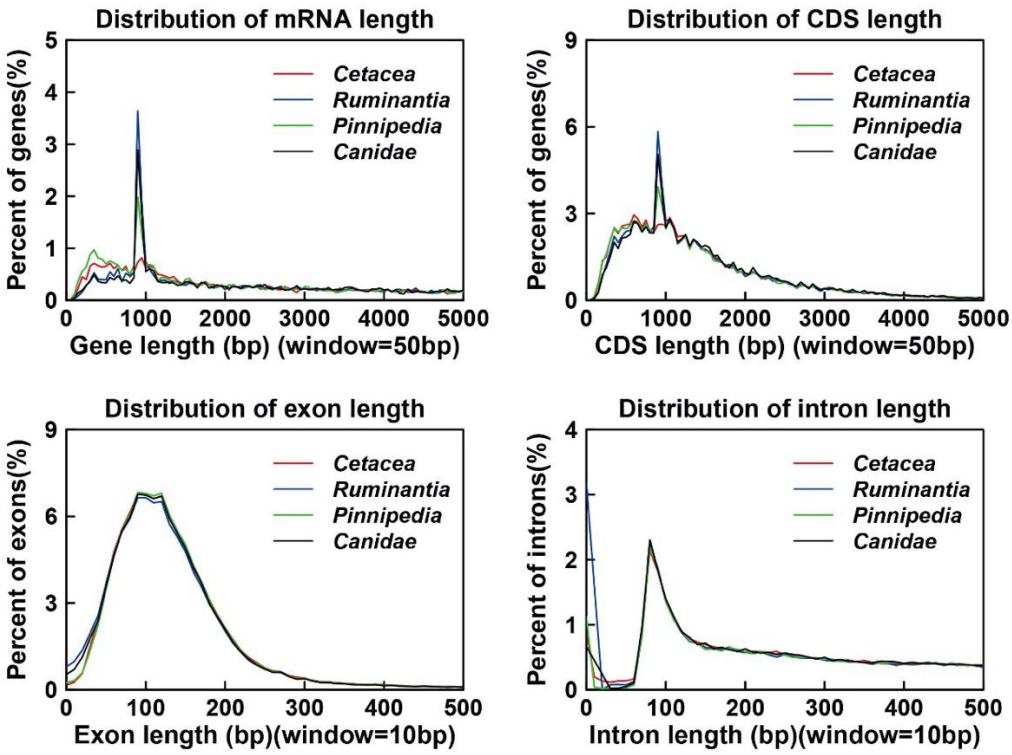


Fig. S3. Examples of the syntenic relationships between marine mammals and their relative terrestrial species. Circos plot of Indo-Pacific bottlenose dolphin vs. cattle, and California sea lion vs. dog. The synteny blocks are linked using lines colored in accordance with different chromosomes.



427

428 **Fig. S4. The distribution of mRNA, CDS, exon, and intron length in**
429 **genomes from cetaceans and pinnipeds, and their terrestrial sister taxa.** Red
430 represents cetaceans, green for pinnipeds, blue for the ruminants and black
431 represents canids.
432

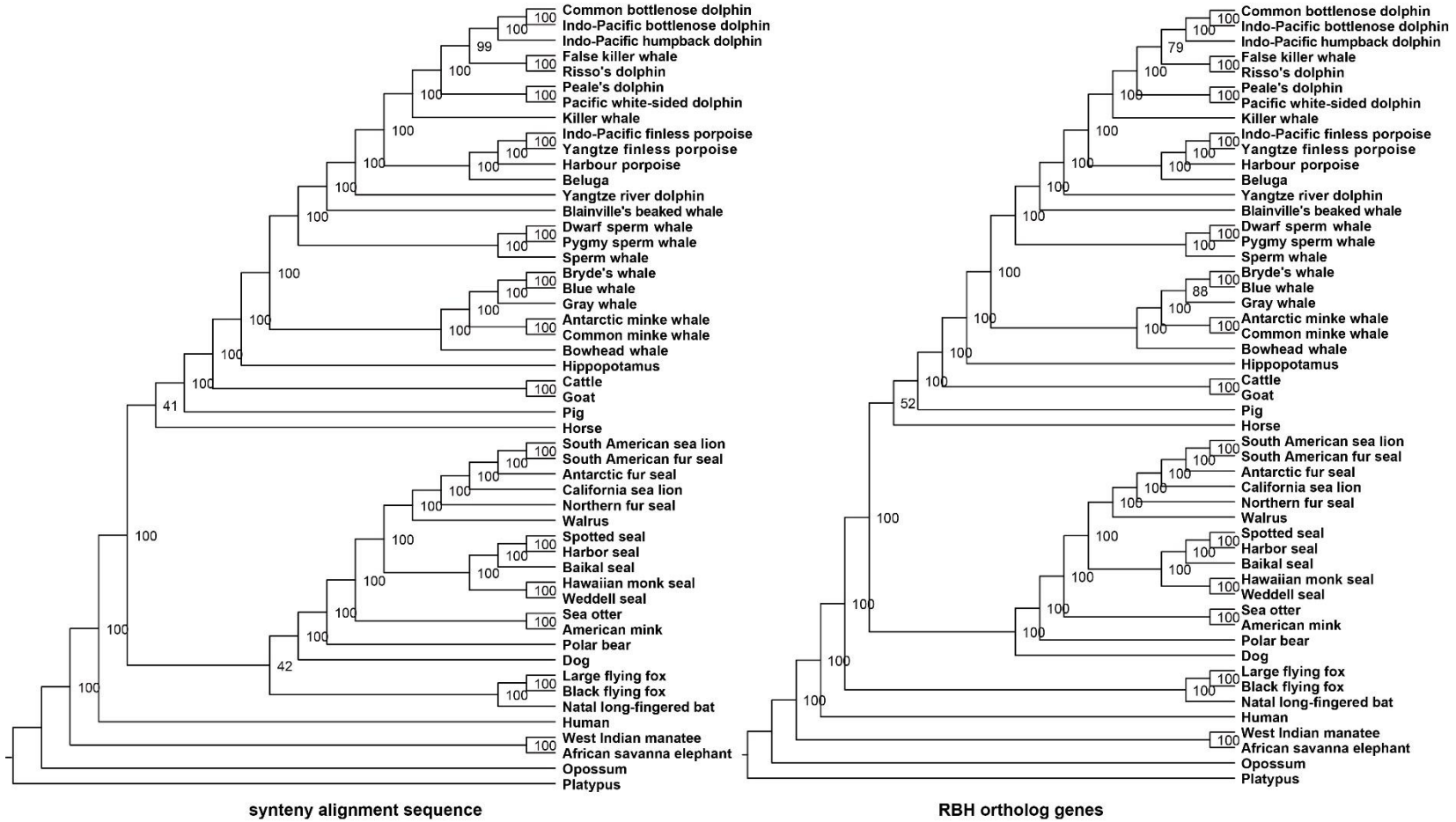


Fig. S5. Species trees inferred by ASTRAL-III obtained using the two different nucleotides datasets.

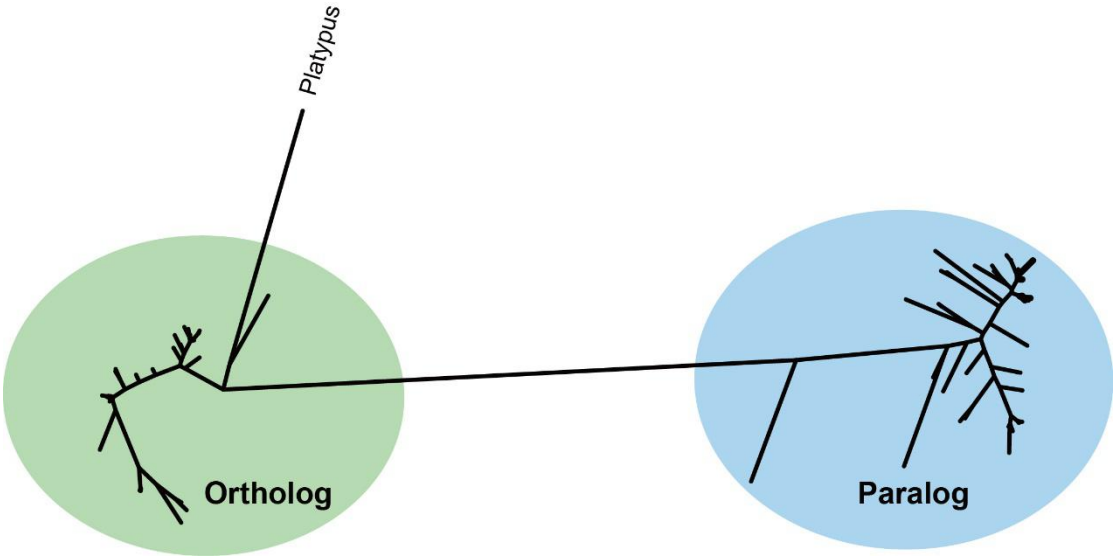


Fig. S6. The highest RF distance homolog sequence inferred tree. The orthologous genes (SCAMP5) are marked in blue and the paralogous genes (SCAMP4) are marked in green, the topology of inferred tree means they are the correct homolog sequences.

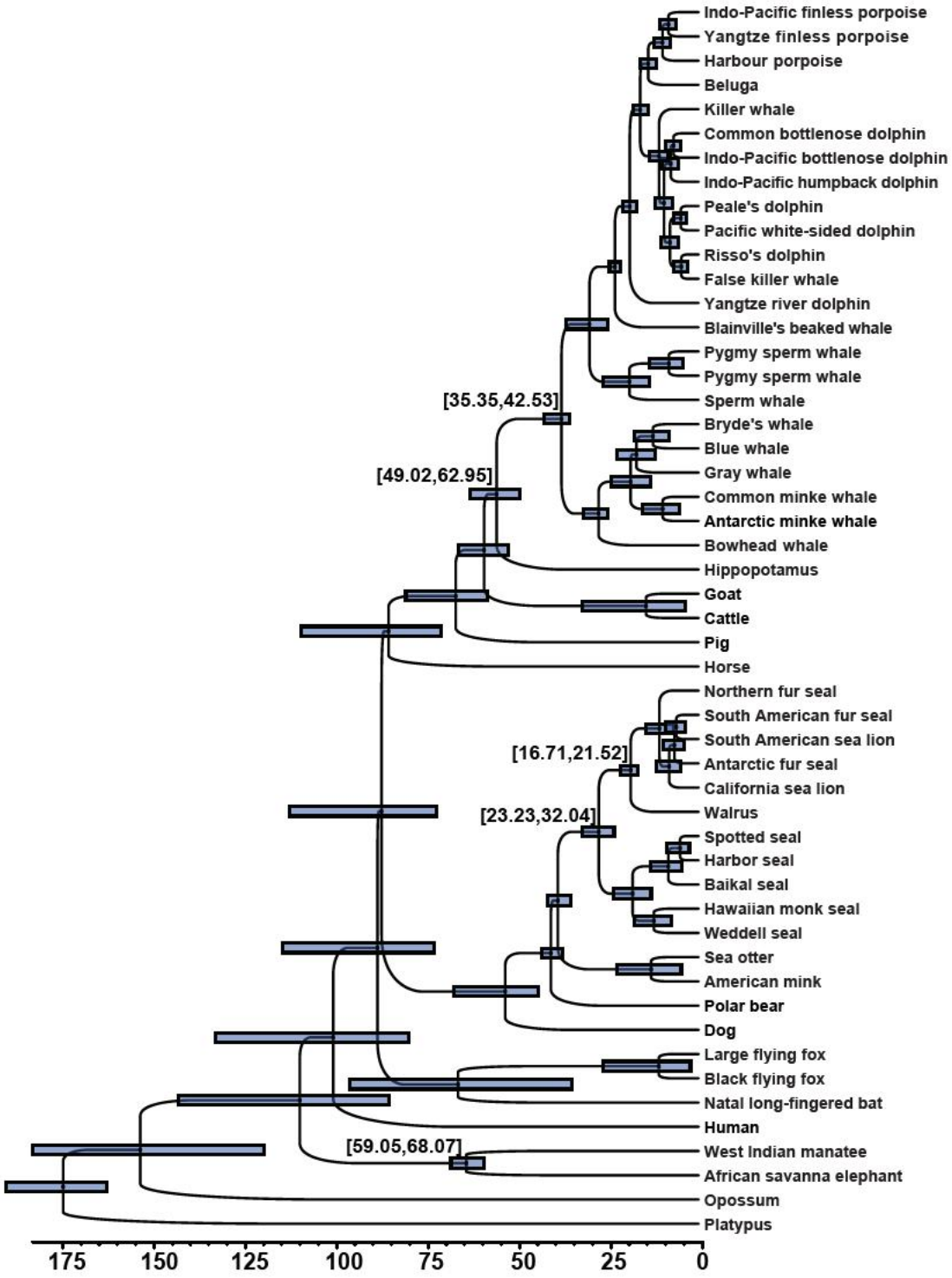
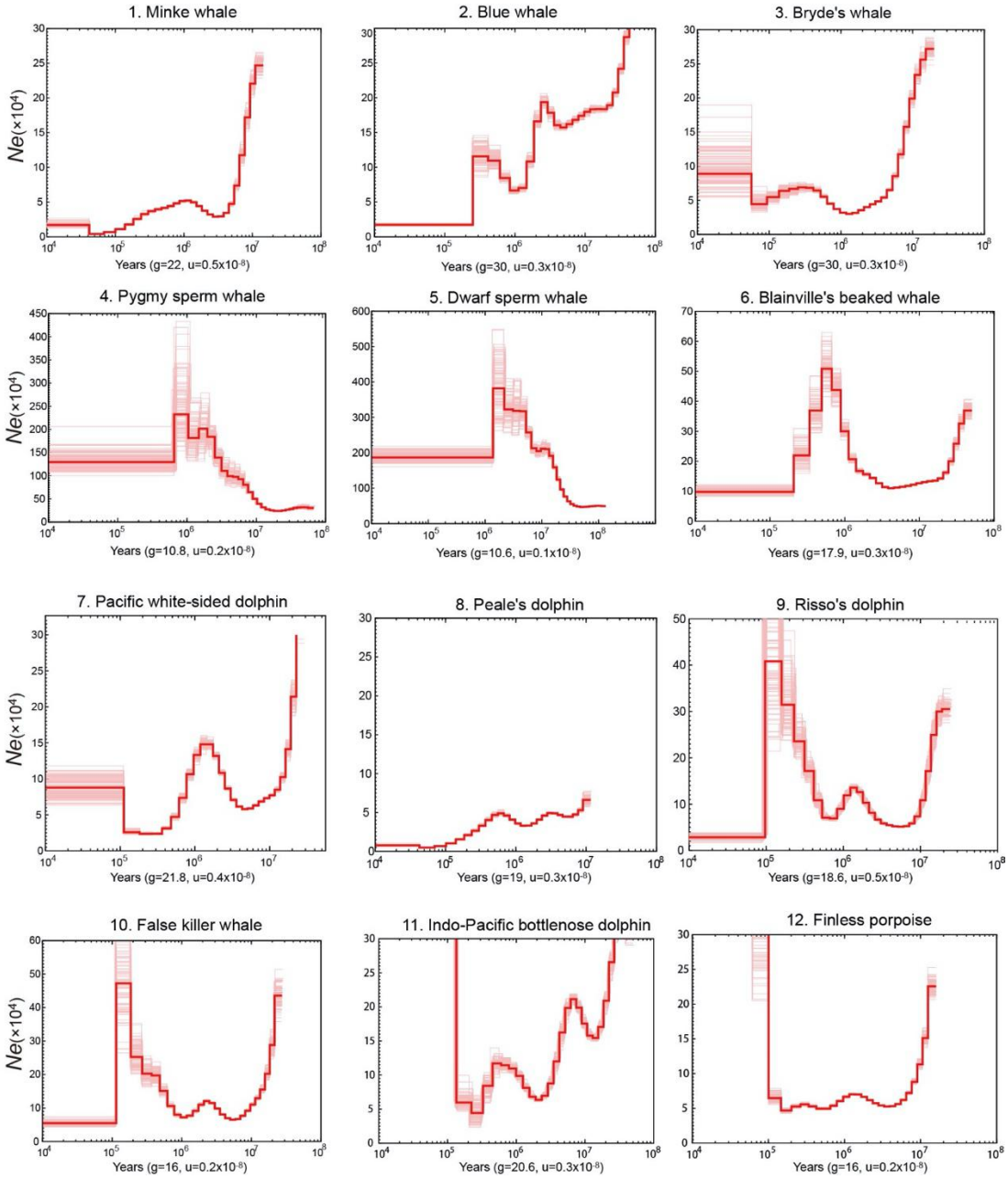


Fig. S7. Divergence time tree of marine mammals estimated from RBH sequence. The estimated divergence time (the unit is Mya) was inferred by MCMCtree, and the blue bars represent 95% confidence intervals.


1. Minke whale
Ne(×10⁴)
Years (g=22, u=0.5x10⁻⁸)
2. Blue whale
Years (g=30, u=0.3x10⁻⁸)
3. Bryde's whale
Years (g=30, u=0.3x10⁻⁸)
4. Pygmy sperm whale
Ne(×10⁴)
Years (g=10.8, u=0.2x10⁻⁸)
5. Dwarf sperm whale
Years (g=10.6, u=0.1x10⁻⁸)
6. Blainville's beaked whale
Years (g=17.9, u=0.3x10⁻⁸)
7. Pacific white-sided dolphin
Ne(×10⁴)
Years (g=21.8, u=0.4x10⁻⁸)
8. Peale's dolphin
Years (g=19, u=0.3x10⁻⁸)
9. Risso's dolphin
Years (g=18.6, u=0.5x10⁻⁸)
10. False killer whale
Ne(×10⁴)
Years (g=16, u=0.2x10⁻⁸)
11. Indo-Pacific bottlenose dolphin
Years (g=20.6, u=0.3x10⁻⁸)
12. Finless porpoise
Years (g=16, u=0.2x10⁻⁸)

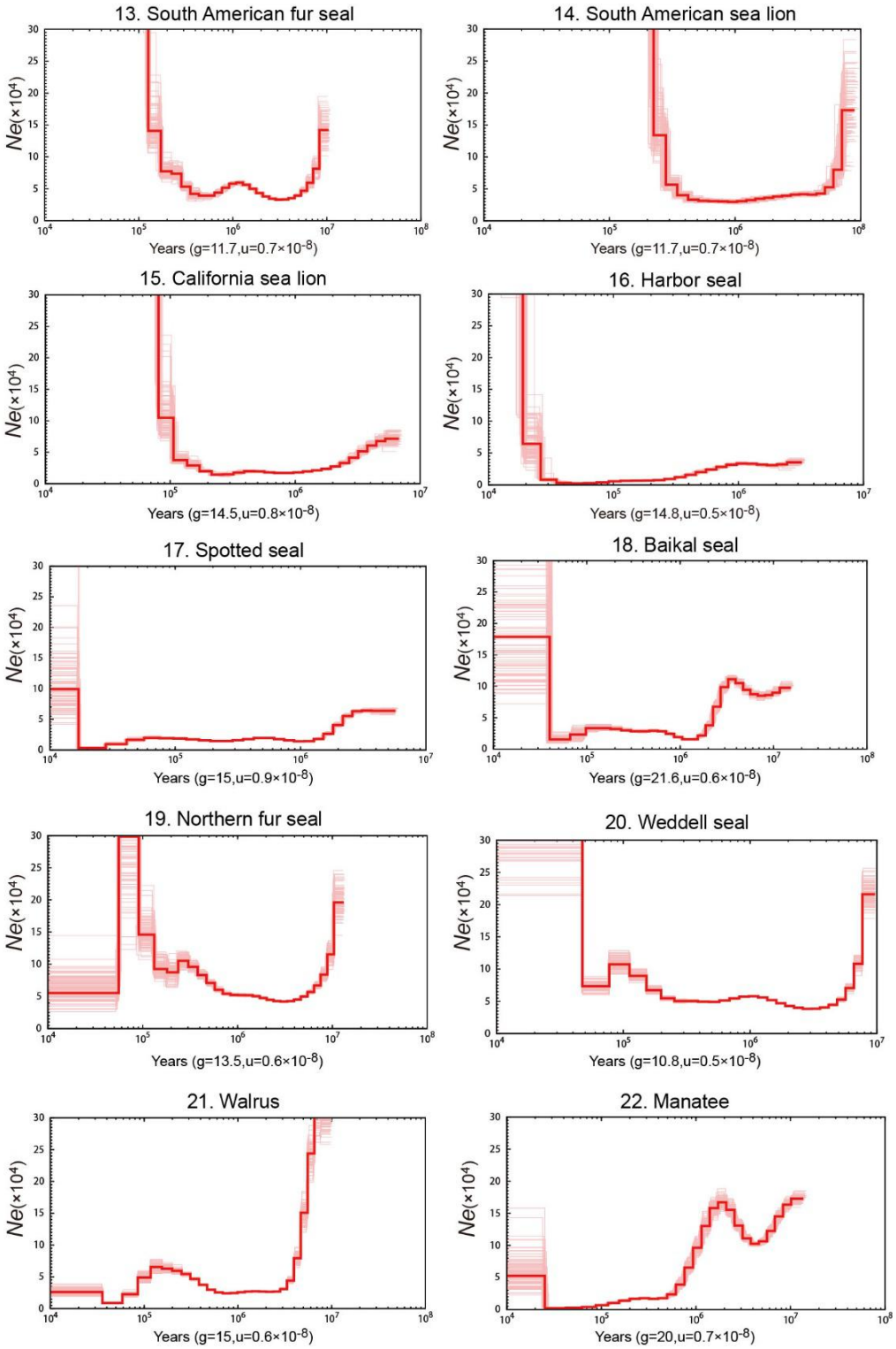


Fig. S8. The effective population size of marine mammals inferred by the PSMC method. Population demographic history of 22 marine mammals are plotted using the pairwise sequential Markovian coalescent method (PSMC), with 100 bootstrap replicates (indicated by lighter lines). Plots were scaled using species-specific mutation rates (μ) and generation times (g).

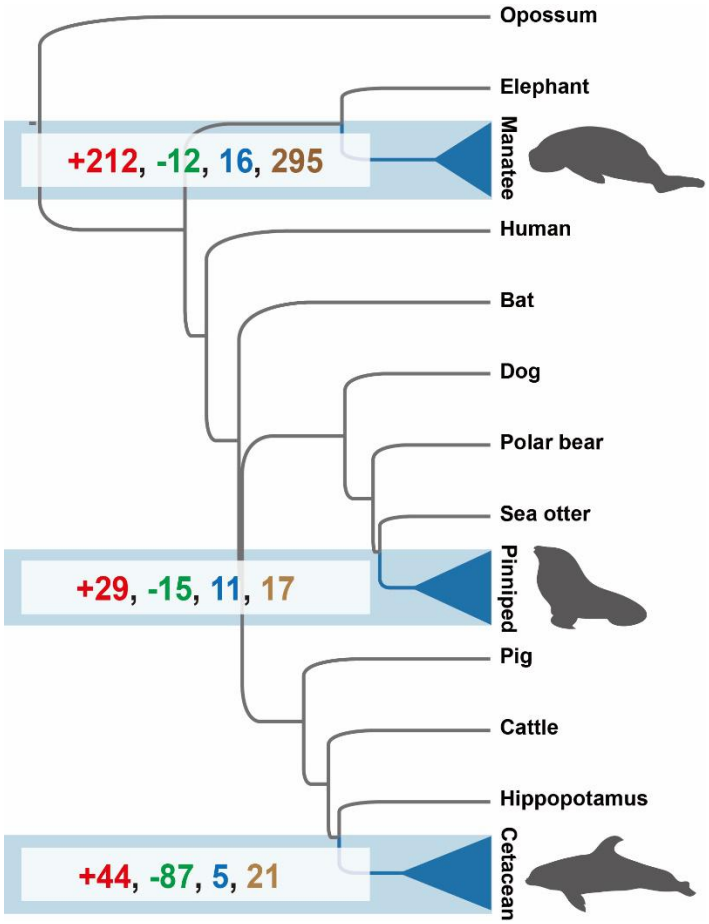


Fig. S9. Evolution of gene families, PSGs and REGs in marine mammals. Expanded gene families (red), contracted gene families (green), PSGs (blue) and REGs (brown) of marine mammals are shown on the phylogenetic tree.

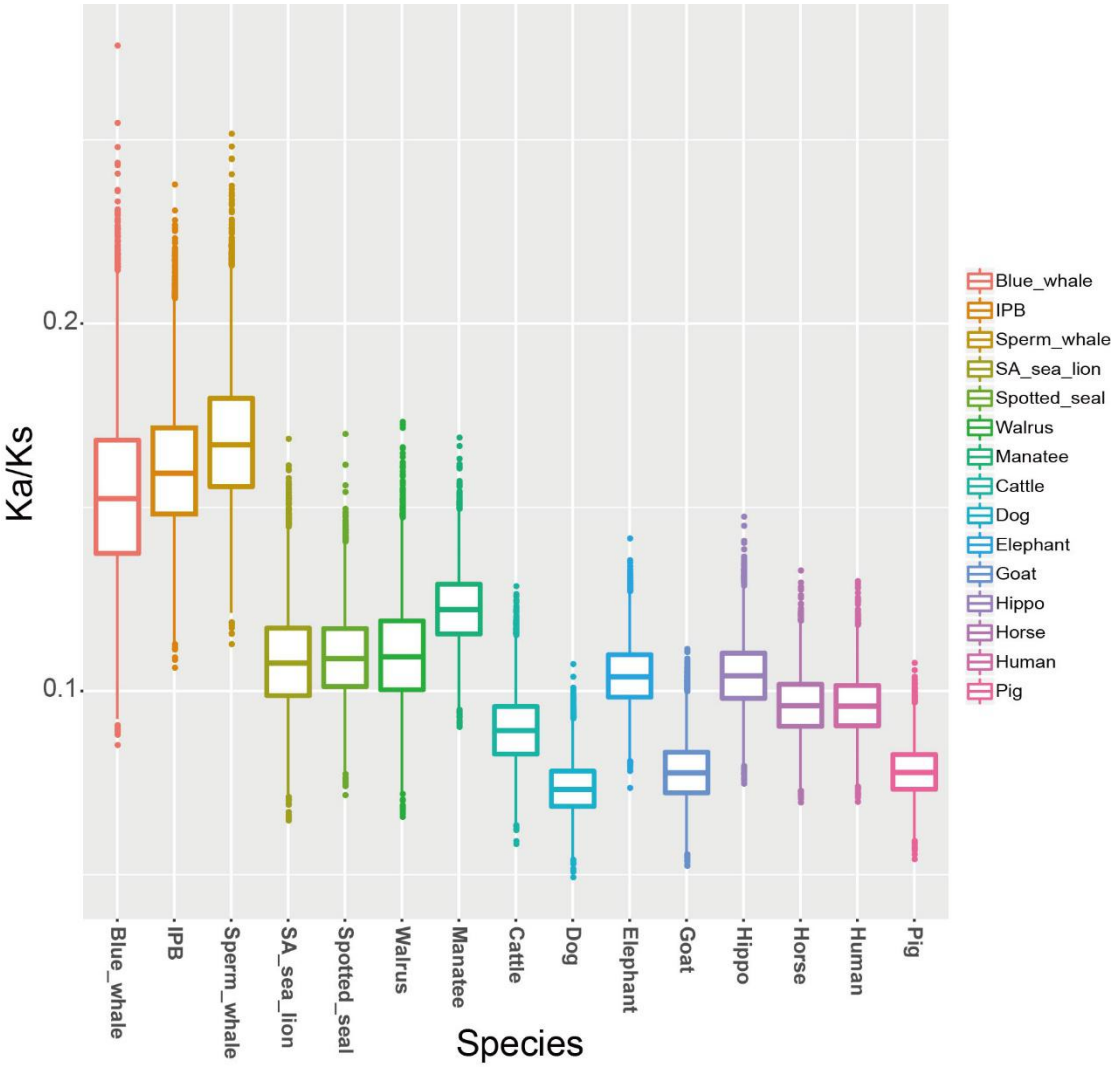


Fig. S10. d_N/d_S values of marine mammalian lineages and their terrestrial relatives. A total of 7,252 orthologous protein-coding genes were used to calculate the d_N/d_S ratios under the PAML free-ratio model. IPB, Indo-Pacific bottlenose dolphin; SA, South American fur seal.

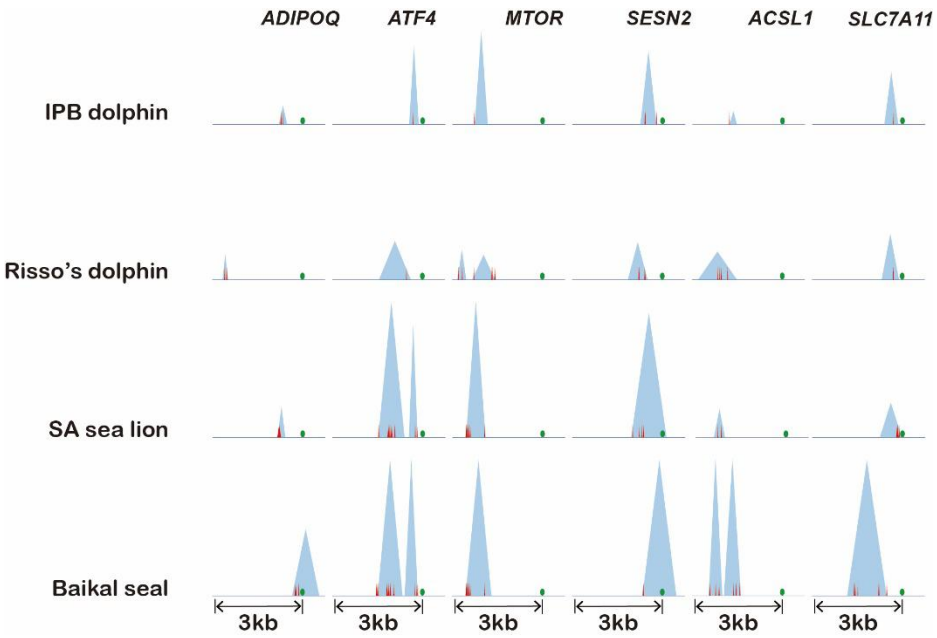


Fig. S11. Example of the ATAC-seq peaks of six candidate CNE-regulated genes in four marine mammals. Green dots, blue peaks and red peaks represent gene start positions, ATAC-seq signal peaks and CNE positions, respectively.

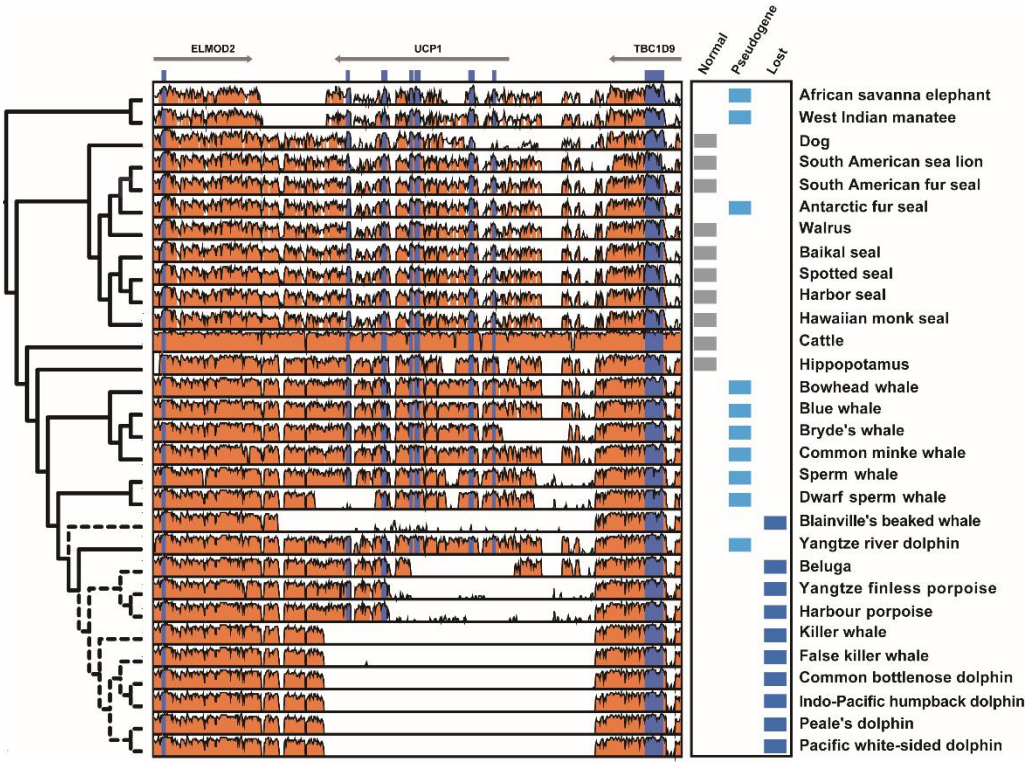
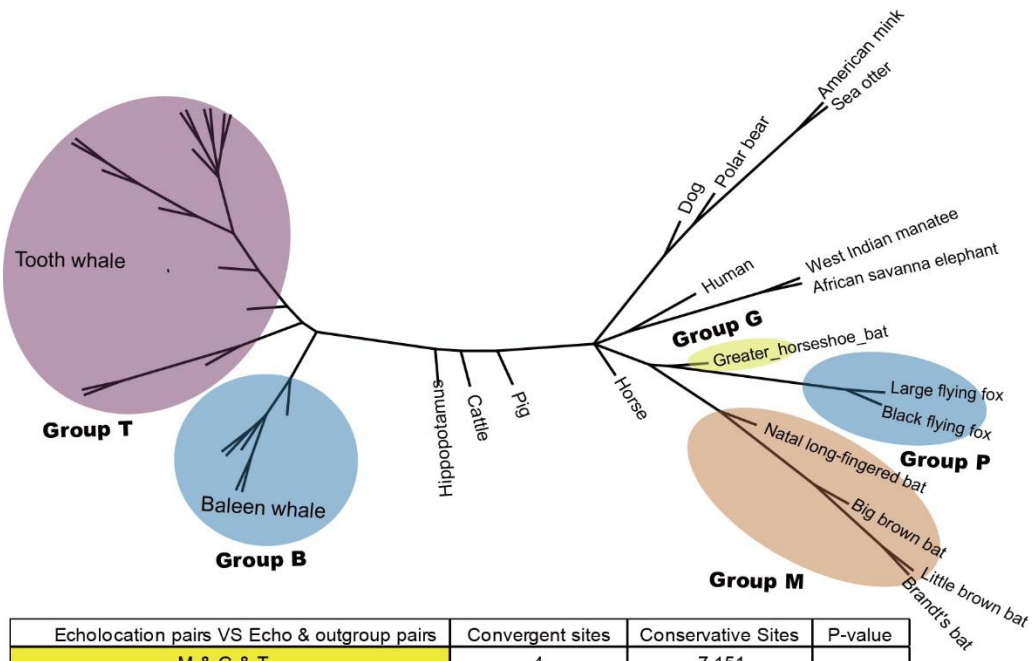


Fig. S12. VISTA sequence conservation plot of the *UCP1* gene and *UCP1* gene status.



Echolocation pairs VS Echo & outgroup pairs	Convergent sites	Conservative Sites	P-value
M & G & T	4	7,151	0.019
M & G & B	0	9,074	
T & M	9	10,317	0.0056
T & P	0	11,068	
G & T	15	10,649	0.031
G & B	6	12,690	
M & G	113	13,117	1.77×10^{-6}
M & P	40	11,072	

Fig. S13. Convergence between echolocating bats and toothed whales. The types for comparison, the number of detected convergent sites, and the number of different pairs of conservative sites are shown on the table.

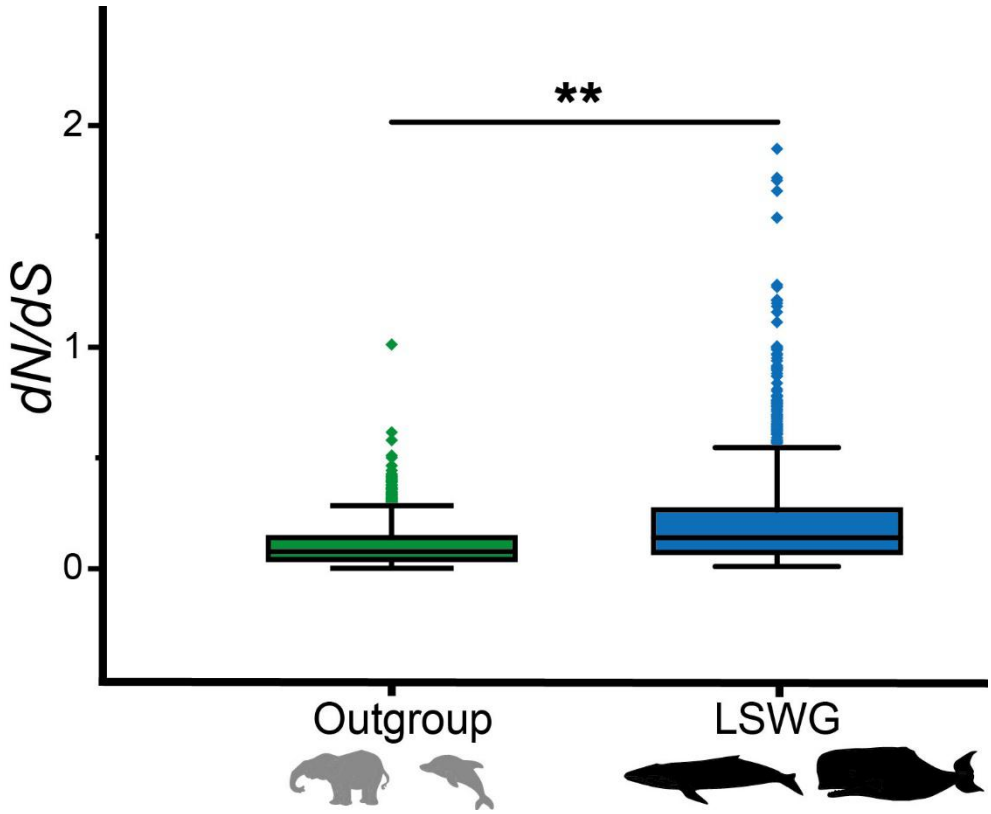


Fig. S14. Comparison of d_N/d_S ratios of a large body size whale group and the background. A total of 1,528 genes related to the body development were retrieved, and the dN/dS ratios calculated under the one-ratio branch model of PAML. ** $P < 0.01$. LSWG denotes large body size group (the giant cetaceans the blue whale and sperm whale).

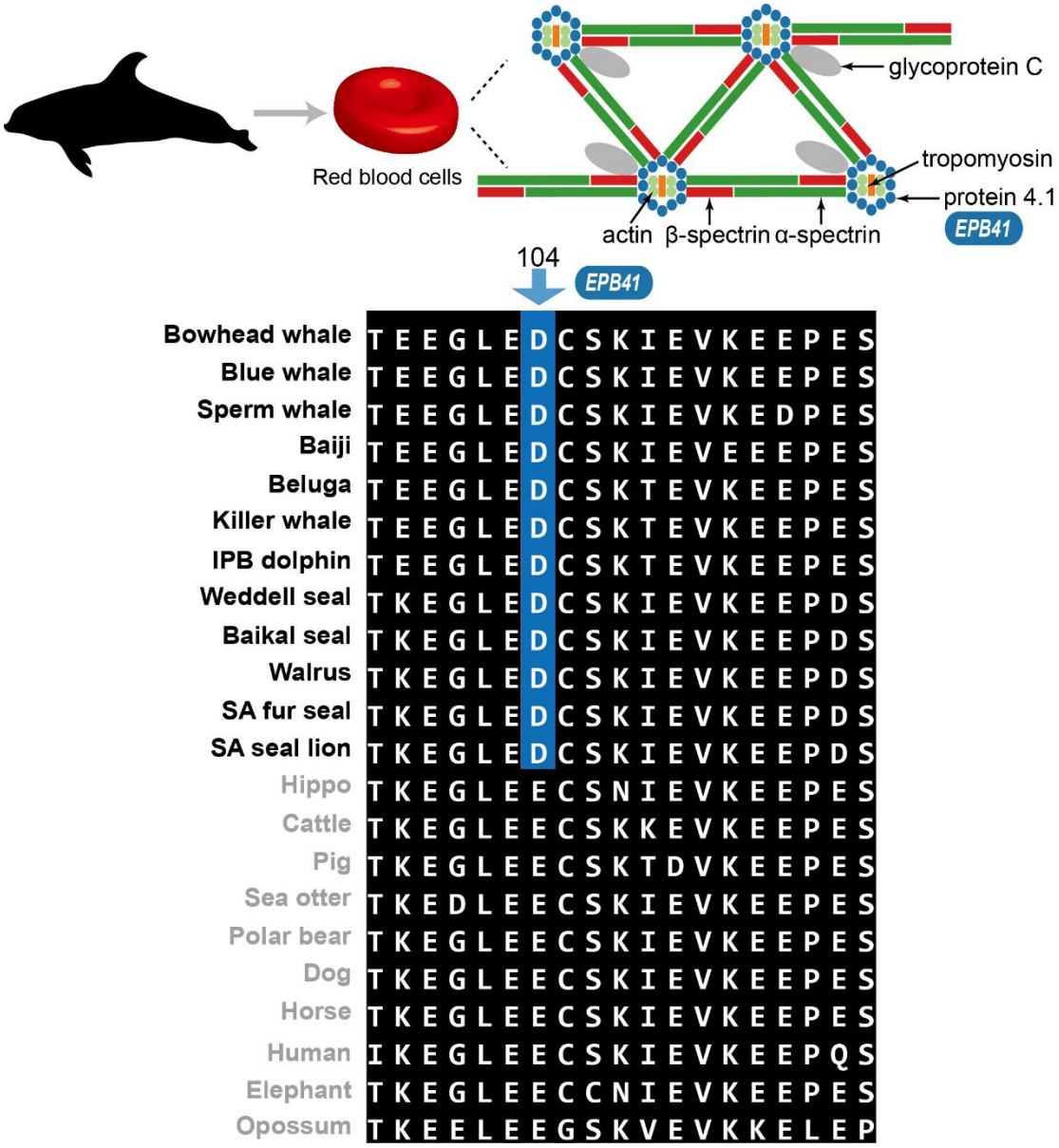


Fig. S15. Multiple sequence alignment showing a shared amino acid change in cetacean and pinniped EPB41.

Supplementary Tables

Table S1. Basic information regarding the sequencing of marine mammals.

Species	Common name	Sample source	Tissue	Sequencing strategy
Arctocephalus australis	South American fur seal	Captive animal in Xiaomeisha Ocean World, Shenzhen, Guangdong, China, being originally from Uruguay	Blood	BGISEQ-500, 10X Genomics, Hi-C
Balaenoptera brydei	Bryde's whale	Stranded animal in Rizhao, Shandong, China	Muscle	HiSeq 2,000, Hi-C
Balaenoptera musculus	Blue whale	Wild animal in Iceland	Skin	10X Genomics
Grampus griseus	Risso's dolphin	Stranded animal in Huizhou, Guangdong, China	Muscle	BGISEQ-500, 10X Genomics, Hi-C
Kogia breviceps	Pygmy sperm whale	Stranded animal in Wenchang, Hainan, China	Muscle	BGISEQ-500, 10X Genomics
Kogia sima	Dwarf sperm whale	Stranded animal in Sanya, Hainan, China	Muscle	BGISEQ-500, 10X Genomics
Lagenorhynchus australis	Peale's dolphin	Wild animal in Chile	Skin	10X Genomics
Lagenorhynchus obliquidens	Pacific white-sided dolphin	Captive animal in Qingdao Polar Ocean World, Qingdao, Shandong, China, being originally from Japan	Blood	BGISEQ-500, 10X Genomics
Mesoplodon densirostris	Blainville's beaked whale	Stranded animal in Ningde, Fujian, China	Muscle	HiSeq 2,000
Neophocaena phocaenoides	Indo-Pacific finless porpoise	Stranded animal in Fuzhou, Fujian, China	Muscle	HiSeq 2,000
Otaria byronia	South American sea lion	Captive animal in Xiaomeisha Ocean World, Shenzhen, Guangdong, China, being originally from Chile	Blood	BGISEQ-500, 10X Genomics, Hi-C
Phoca largha	Spotted seal	Captive animal in Xiaomeisha Ocean Word, Shenzhen, Guangdong, China, being originally from China	Blood	BGISEQ-500, 10X Genomics, Hi-C
Phoca vitulina	Harbor seal	Captive animal in Qingdao Polar Ocean World, Qingdao, Shandong, China, being originally from Japan	Blood	BGISEQ-500, 10X Genomics

Pseudorca crassidens	False killer whale	Captive animal in Qingdao Polar Ocean World, Qingdao, Shandong, China, being originally from Japan	Blood	BGISEQ-500, 10X Genomics
Pusa sibirica	Baikal seal	Captive animal in Qingdao Polar Ocean World, Qingdao, Shandong, China, being originally from Baikal, Russia	Muscle	BGISEQ-500, 10X Genomics, Hi-C
Tursiops aduncus	Indo-Pacific bottlenose dolphin	Captive animal in Xiaomeisha Ocean World, Shenzhen, Guangdong, China, being originally from Indonesia	Blood	BGISEQ-500, 10X Genomics, Hi-C
Zalophus californianus	California sea lion	Captive animal in Chengdu Polar Haichang Ocean Park, Chengdu, Sichuan, China, being originally from California, USA	Muscle	BGISEQ-500, 10X Genomics, Hi-C

Table S2. Summary of the marine mammals raw sequencing data generated in this study.

Species	Common name	WGS (Gb, BGISEQ-500)	WGS (Gb, Hiseq)	10X (Gb)	Hi-C (Gb)
Kogia sima	Dwarf sperm whale	279	/	393	/
Kogia breviceps	Pygmy sperm whale	145	/	152	/
Pseudorca crassidens	False killer whale	249	/	350	/
Lagenorhynchus obliquidens	Pacific white-sided dolphin	290	/	409	/
Grampus griseus	Risso's dolphin	270	/	265	147
Tursiops aduncus	Indo-Pacific bottlenose dolphin	280	/	280	115
Zalophus californianus	California sea lion	268	/	367	118
Otaria byronia	South American sea lion	242	/	264	120
Arctocephalus australis	South American fur seal	267	/	296	120
Phoca vitulina	Harbor seal	304	/	406	/
Phoca largha	Spotted seal	276	/	406	134
Pusa sibirica	Baikal seal	261	/	379	135
Lagenorhynchus australis	Peale's dolphin	/	/	281	/
Balaenoptera musculus	Blue whale	/	/	270	/
Balaenoptera brydei	Bryde's whale	/	631	/	242
Mesoplodon densirostris	Blainville's beaked whale	/	493	/	/
Neophocaena phocaenoides	Indo-Pacific finless porpoise	/	506	/	/

Table S3. Software and tools used for genome assembly.

Species	Common name	Assembly software
Arctocephalus australis	South American fur seal	Supernova+KGF+Gapcloser+ARKS+3d-DNA
Balaenoptera brydei	Bryde's whale	Platanus
Balaenoptera musculus	Blue whale	Supernova+ARKS
Grampus griseus	Risso's dolphin	Supernova+KGF+Gapcloser+3d-DNA
Kogia breviceps	Pygmy sperm whale	Supernova+KGF+Gapcloser+ARKS
Kogia sima	Dwarf sperm whale	Supernova+KGF+Gapcloser+ARKS
Lagenorhynchus australis	Peale's dolphin	Supernova+KGF+Gapcloser
Lagenorhynchus obliquidens	Pacific white-sided dolphin	Supernova+KGF+Gapcloser
Mesoplodon densirostris	Blainville's beaked whale	Platanus
Neophocaena phocaenoides	Indo-Pacific finless porpoise	Platanus
Otaria byronia	South American sea lion	Supernova+KGF+Gapcloser+3d-DNA
Phoca largha	Spotted seal	Supernova+KGF+Gapcloser+ARKS+3d-DNA
Phoca vitulina	Harbor seal	Supernova+KGF+Gapcloser+ARKS
Pseudorca crassidens	False killer whale	Supernova+KGF+Gapcloser+ARKS
Pusa sibirica	Baikal seal	Supernova+KGF+Gapcloser+ARKS+3d-DNA
Tursiops aduncus	Indo-Pacific bottlenose dolphin	Supernova+KGF+Gapcloser+3d-DNA
Zalophus californianus	California sea lion	Supernova+KGF+Gapcloser+ARKS+3d-DNA

Table S4. Estimated genome sizes by using k-mer analysis.

Species	Common name	Assembled size (Gb)	Estimated size (Gb)
Arctocephalus australis	South American fur seal	2.49	2.31
Balaenoptera brydei	Bryde's whale	2.37	2.11
Balaenoptera musculus	Blue whale	2.43	2.45
Grampus griseus	Risso's dolphin	2.66	2.47
Kogia breviceps	Pygmy sperm whale	2.56	2.47
Kogia sima	Dwarf sperm whale	2.59	2.47
Lagenorhynchus australis	Peale's dolphin	2.39	2.46
Lagenorhynchus obliquidens	Pacific white-sided dolphin	2.46	2.47
Mesoplodon densirostris	Blainville's beaked whale	2.47	2.08
Neophocaena phocaenoides	Indo-Pacific finless porpoise	2.62	2.81
Otaria byronia	South American sea lion	2.45	2.51
Phoca largha	Spotted seal	2.39	2.29
Phoca vitulina	Harbor seal	2.39	2.34
Pseudorca crassidens	False killer whale	2.43	2.55
Pusa sibirica	Baikal seal	2.43	2.33
Tursiops aduncus	Indo-Pacific bottlenose dolphin	2.41	2.55
Zalophus californianus	California sea lion	2.48	2.46

Table S5. Previously published genomes of the marine mammal orders Cetacea, Pinnipedia, and Sirenia.

Order	Family	Species	Common name	Version	Assembled genome size (Gb)
Mysticeti	Balaenidae	*Balaena mysticetus*	**Bowhead whale**	/	2.31
	Rorquals	*Balaenoptera acutorostrata*	**Common minke whale**	BalAcu1.0	2.43
		Balaenoptera bonaerensis	**Antarctic minke whale**	ASM97880v1	2.23
	Eschrichtiidae	*Eschrichtius robustus*	**Gray whale**	ASM218922v1	2.85
	Physeteroidea	*Physeter macrocephalus*	**Sperm whale**	ASM283717v2	2.58
Odontoceti	Monodontidae	*Delphinapterus leucas*	**Beluga**	ASM228892v2	2.35
	Delphinidae	*Orcinus orca*	**Killer whale**	Oorc_1.1	2.37
		Sousa chinensis	**Indo-Pacific humpback dolphin**	/	2.46
		Tursiops truncatus	**Common bottlenose dolphin**	Tur_tru_v1	2.13
	Phocoenidae	*Neophocaena asiaeorientalis*	**Yangtze finless porpoise**	Neophocaena_asiaeorientalis_v1	2.28
		Phocoena phocoena	**Harbor porpoise**	ASM307100v1	2.44
	Lipotidae	*Lipotes vexillifer*	**Yangtze river dolphin**	Lipotes_vexillifer_v1	2.42
Pinnipedia	Otariidae	*Callorhinus ursinus*	**Northern fur seal**	ASM326570v1	2.71
		Arctocephalus gazella	**Antarctic fur seal**	ArcGazv1.4	2.31
	Odobenidae	*Odobenus rosmarus*	**Walrus**	Oros_1.0	2.40
	Phocidae	*Neomonachus schauinslandi*	**Hawaiian monk seal**	ASM220157v1	2.40
		Leptonychotes weddellii	**Weddell seal**	LepWed1.0	3.16
Sirenia	Trichechidae	*Trichechus manatus*	**West Indian manatee**	TriManLat1.0	3.10

Table S6. BUSCO complete ratios for the 17 marine mammal genome assemblies generated in this study.

Species	Common name	Ratios (%)
Arctocephalus australis	South American fur seal	93.0
Balaenoptera brydei	Bryde's whale	87.3
Balaenoptera musculus	Blue whale	94.4
Grampus griseus	Risso's dolphin	90.8
Kogia breviceps	Pygmy sperm whale	87.6
Kogia sima	Dwarf sperm whale	90.0
Lagenorhynchus australis	Peale's dolphin	87.9
Lagenorhynchus obliquidens	Pacific white-sided dolphin	94.0
Mesoplodon densirostris	Blainville's beaked whale	87.1
Neophocaena phocaenoides	Indo-Pacific finless porpoise	81.2
Otaria byronia	South American sea lion	93.8
Phoca largha	Spotted seal	95.2
Phoca vitulina	Harbor seal	95.4
Pseudorca crassidens	False killer whale	93.7
Pusa sibirica	Baikal seal	92.4
Tursiops aduncus	Indo-Pacific bottlenose dolphin	90.8
Zalophus californianus	California sea lion	92.1

Table S7. Transposable elements in the marine mammal genomes (%).

Species	Common name	DNA	LINE	SINE	LTR	Other	Total
Arctocephalus australis	South American fur seal	1.488158	31.439661	0.535981	8.796351	0.065415	35.598588
Balaenoptera brydei	Bryde's whale	1.688984	43.61634	0.429382	7.0336	0.004828	47.2219
Balaenoptera musculus	Blue whale	1.307166	32.745222	0.396537	6.722025	0.06616	35.844798
Grampus griseus	Risso's dolphin	1.461934	39.320703	0.450959	6.541915	0.035193	42.539802
Kogia breviceps	Pygmy sperm whale	1.79026	28.787559	0.53144	6.626339	0.069728	32.976348
Kogia sima	Dwarf sperm whale	1.623577	34.571321	0.594562	7.470587	0.081442	38.560482
Lagenorhynchus australis	Peale's dolphin	1.581625	35.094537	0.563309	7.789377	0.051763	38.81654
Lagenorhynchus obliquidens	Pacific white-sided dolphin	1.408483	36.563296	0.475404	7.127017	0.000466	40.181447
Mesoplodon densirostris	Blainville's beaked whale	1.755442	41.26843	0.5808	7.813376	0.160903	45.98339
Neophocaena phocaenoides	Indo-Pacific finless porpoise	1.967851	37.23986	0.7664	5.963169	0.067085	40.99384
Otaria byronia	South American sea lion	1.333276	36.737434	0.483567	7.627098	0.095502	40.427213
Phoca largha	Spotted seal	1.507089	35.488439	0.530881	3.165685	5.717489	39.727357
Phoca vitulina	Harbor seal	1.598398	31.592228	0.578285	7.515027	0.772087	35.6998
Pseudorca crassidens	False killer whale	1.33549	34.509515	0.421458	7.471515	0.003603	37.825734
Pusa sibirica	Baikal seal	1.562022	31.713937	0.553942	8.077584	0.016157	35.81291
Tursiops aduncus	Indo-Pacific bottlenose dolphin	1.448769	36.299031	0.488658	2.993166	0.012115	38.537753
Zalophus californianus	California sea lion	1.455071	31.844194	0.529183	8.158264	0.034161	35.73657

Table S8. Average gene, CDS length and exon numbers of annotated marine mammal genomes.

Species	Common name	Average gene length (bp)	Average CDS length (bp)	Average exon number
Arctocephalus australis	South American fur seal	36594.16	1602.21	8.7
Balaenoptera brydei	Bryde's whale	40202.26	1675.59	9.29
Balaenoptera musculus	Blue whale	38477.15	1634.99	9.44
Grampus griseus	Risso's dolphin	37253.3	1682	9.55
Kogia breviceps	Pygmy sperm whale	41167.06	1682	9.97
Kogia sima	Dwarf sperm whale	35330.79	1590.44	8.48
Lagenorhynchus australis	Peale's dolphin	36497.67	1613.97	9.51
Lagenorhynchus obliquidens	Pacific white-sided dolphin	39031.04	1647.9	9.51
Mesoplodon densirostris	Blainville's beaked whale	39009.86	1568.7	8.34
Neophocaena phocaenoides	Indo-Pacific finless porpoise	48045.64	1615.69	10
Otaria byronia	South American sea lion	34534.18	1648.37	9.23
Phoca largha	Spotted seal	40748.23	1687.55	9.42
Phoca vitulina	Harbor seal	36290.65	1615.14	8.79
Pseudorca crassidens	False killer whale	39191.76	1650.82	9.54
Pusa sibirica	Baikal seal	36762.16	1648.01	8.88
Tursiops aduncus	Indo-Pacific bottlenose dolphin	37268.14	1590.45	8.88
Zalophus californianus	California sea lion	34837.15	1548.84	8.92

Table S9. Outgroup species.

Species	Common name	Version
Bos taurus	Cattle	ARS-UCD1.2
Canis lupus familiaris	Dog	CanFam3.1
Capra hircus	Goat	ARS1
Enhydra lutris	Sea otter	ASM228890v2
Eptesicus fuscus	Big brown bat	Mnat.v1EptFus1.0
Equus caballus	Horse	EquCab3.0
Hippopotamus amphibius	Hippopotamus	ASM299558v1
Homo sapiens	Human	GRCh38.p12
Loxodonta africana	African savanna elephant	Loxafr3.0
Miniopterus natalensis	Natal long-fingered bat	Mnat.v1
Monodelphis domestica	Opossum	MonDom5
Myotis brandtii	Brandt's bat	ASM41265v1
Myotis lucifugus	Little brown bat	Myoluc2.0
Neovison vison	American mink	NNQGG.v01
Ornithorhynchus anatinus	Platypus	Ornithorhynchus_anatinus-5.0.1
Pteropus alecto	Black flying fox	ASM32557v1
Pteropus vampyrus	Large flying fox	Pvam_2.0
Rhinolophus ferrumequinum	Greater horseshoe bat	ASM46549v1
Sus scrofa	Pig	Sscrofa11.1
Ursus maritimus	Polar bear	UrsMar_1.0

Table S10. Mutation rate and generation times used in the PSMC analyses.

Species	Common name	Mutation rate	Generation time (year)
Balaena mysticetus	Bowhead whale	9.1017×10^{-5}	52
Balaenoptera acutorostrata	Common minke whale	2.2467×10^{-4}	22
Balaenoptera brydei	Bryde's whale	1.7044×10^{-4}	18
Balaenoptera musculus	Blue whale	1.0874×10^{-4}	31
Eschrichtius robustus	Gray whale	2.0389×10^{-4}	19.3
Physeter macrocephalus	Sperm whale	1.7734×10^{-3}	27.5
Kogia sima	Dwarf sperm whale	9.5867×10^{-5}	10.6
Kogia breviceps	Pygmy sperm whale	2.0178×10^{-4}	10.8
Delphinapterus leucas	Beluga whale	1.7214×10^{-4}	16
Orcinus orca	Killer whale	4.0659×10^{-4}	27
Tursiops truncatus	Common bottlenose dolphin	3.3217×10^{-4}	20
Sousa chinensis	Indo-Pacific Humpbacked Dolphin	2.689×10^{-4}	19.8
Lagenorhynchus obliquidens	Pacific white-sided dolphin	1.6132×10^{-4}	21.8
Lagenorhynchus australis	Peale's dolphin	1.7530×10^{-4}	19
Grampus griseus	Risso's dolphin	2.7463×10^{-4}	18.6
Tursiops aduncus	Indo-Pacific bottlenose dolphin	1.3844×10^{-4}	20.6
Pseudorca crassidens	False killer whale	1.4077×10^{-4}	16
Neophocaena asiaeorientalis	Yangtze finless porpoise	3.3681×10^{-4}	12
Neophocaena phocaenoides	Indo-Pacific finless porpoise	2.9354×10^{-4}	14.4
Mesoplodon densirostris	Blainville's beaked whale	1.5975×10^{-4}	17.9
Callorhinus ursinus	Northern fur seal	4.6394×10^{-4}	13.5
Zalophus californianus	California sea lion	5.317×10^{-4}	14.5
Otaria byronia	South American sea lion	6.6409×10^{-4}	10.6
Arctocephalus australis	South American fur seal	5.5808×10^{-4}	11.7
Odobenus rosmarus	Walrus	3.8167×10^{-4}	15
Leptonychotes weddellii	Weddell seal	4.9391×10^{-4}	10.8
Phoca vitulina	Harbor seal	3.3845×10^{-4}	14.8
Phoca largha	Spotted seal	6.0324×10^{-4}	15
Pusa sibirica	Baikal seal	2.8464×10^{-4}	21.6
Trichechus manatus	West Indian manatee	3.5758×10^{-4}	20

Table S11. Heterozygosity information of marine mammals.

Family	Species	Heterozygosity	Average
Balaenopteridae	*Balaenoptera musculus*	2.44E-03	1.51E-03
	Balaenoptera brydei	8.30E-04	
	Balaenoptera acutorostrata	1.26E-03	
Balaenidae	*Balaena mysticetus*	2.73E-03	2.73E-03
Physeteroidea	*Kogia sima*	5.55E-03	4.43E-03
	Kogia breviceps	6.46E-03	
	Physeter macrocephalus	1.29E-03	
Ziphiidae	*Mesoplodon densirostris*	2.08E-03	4.43E-03
Lipotidae	*Lipotes vexillifer*	1.21E-04	1.21E-04
Phocoenidae	*Neophocaena asiaeorientalis*	1.26E-03	1.50E-03
	Neophocaena phocaenoides	1.74E-03	
Delphinnidae	*Grampus griseus*	1.49E-03	1.05E-03
	Orcinus orca	7.20E-04	
	Lagenorhynchus australis	5.84E-04	
	Lagenorhynchus obliquidens	1.26E-03	
	Pseudorca crassidens	1.07E-03	
	Tursiops aduncus	1.59E-03	
	Sousa chinensis	6.62E-04	
Phocidae	*Leptonychotes weddellii*	1.50E-03	9.11E-04
	Phoca largha	8.28E-04	
	Phoca vitulina	3.47E-04	
	Pusa sibirica	9.68E-04	
Odobenidae	*Odobenus rosmarus*	1.03E-03	1.03E-03
Otariidae	*Arctocephalus australis*	1.34E-03	1.11E-03
	Callorhinus ursinus	1.67E-03	
	Otaria byronia	8.18E-04	
	Zalophus californianus	6.22E-04	
Trichechidae	*Trichechus manatus*	1.55E-03	1.55E-03

Table S13. Large deletions in Indo-Pacific bottlenose (IPB) dolphin genome compared to cattle genome.

Chromosome IPB dolphin	Deletion point	Chromosome of cattle	Start	End
chr_2	129,910,262	chr1	148,490,253	153,231,467
chr_5	4,140,003	chr4	50,387,276	75,171,314
chr_8	31,004,836	chr5	25,359,997	42,803,076
chr_8	31,004,836	chr5	51,732,212	60,311,954
chr_20	108,230,050	chr7	39,169,909	82,843,312
chr_15	81,874,016	chr8	66,049,394	76,045,636
chr_15	81,874,016	chr8	85,472,271	89,071,912
chr_10	63,022,590	chr12	70,302,185	74,304,154
chr_17	28,019,333	chr13	53,887,632	60,223,925
chr_19	217,859	chr18	342,448	14,345,208
chr_18	42,638,219	chr19	14,959,883	36,083,419

Table S14. Large insertions in Indo-Pacific bottlenose (IPB) dolphin genome compared to cattle genome.

Chromosome of IPB dolphin	Start	End
chr_5	51,340,049	74,019,070
chr_8	42,201,330	77,840,594
chr_15	10,830,011	20,765,705

Table S15. Confirmation of three large insertions in IPB dolphin.

Insertion Region	Species	Corresponding regions (chromosome: start-end)
chr8:42201330-77840594	Sperm whale	chr6:40437811-52401263;chr6:75178781-99882195
chr_5:51340049-74019070		chr5:32359267-55689733
chr_15:10830011-20765705		chr9:87535423-97463036
chr8:42201330-77840594	Bowhead whale	scaffold_22:1-3193872;scaffold_19:142-3335931;scaffold_118
chr_5:51340049-74019070		scaffold_52:76-2539998;scaffold_69:137-2441457;scaffold_10
chr_15:10830011-20765705		scaffold_18:1-3366608;scaffold_180:1024-1775262;scaffold_1
chr8:42201330-77840594	Blue whale	scaffold95:1645-7650949;scaffold287:1-4637484;scaffold106:
chr_5:51340049-74019070		scaffold11:6926-15252381;scaffold70:657-7634658;
chr_15:10830011-20765705		scaffold127:1-4014901;scaffold210:9075-2756208;scaffold119

chr8:42201330-77840594	Pygmy sperm whale	scaffold75:1-7617426;scaffold105:5992-4651835;scaffold37:1
chr_5:51340049-74019070		scaffold5:25139744-47779888;
chr_15:10830011-20765705		scaffold41:211419-7287506;scaffold61:7844873-10566195;
chr8:42201330-77840594	White whale	NW_019160865.1:13177876-29521609;NW_019160911.1:419
chr_5:51340049-74019070		NW_019160856.1:31769204-54493892;
chr_15:10830011-20765705		NW_019160906.1:10873544-14841884;NW_019160998.1:414
chr8:42201330-77840594	Harbor porpoise	NW_020185339.1:22174856-33178146;NW_020172451.1:138 208820;NW_020172792.1:271-407320;
chr_5:51340049-74019070		NW_020174149.1:5537-8553837;NW_020174561.1:7187946-
chr_15:10830011-20765705		NW_020173550.1:1250213-5403958;NW_020175675.1:19912

Table S16. KEGG enrichment of significantly contracted gene families in marine mammals.

Groups	ID	Description	p-value	Adjusted p-value
Cetacean	ko04740	Olfactory transduction	8.9286E-287	3.0089E-284
	ko05320	Autoimmune thyroid disease	4.28465E-32	1.44393E-29
	ko00053	Ascorbate and aldarate metabolism	3.08701E-26	1.04032E-23
	ko04650	Natural killer cell mediated cytotoxicity	5.82101E-22	1.96168E-19
	ko04622	RIG-I-like receptor signaling pathway	1.50421E-20	5.06917E-18
	ko00040	Pentose and glucuronate interconversions	2.01267E-19	6.7827E-17
	ko00860	Porphyrin and chlorophyll metabolism	1.67898E-13	5.65815E-11
	ko00982	Drug metabolism - cytochrome P450	5.721E-13	1.92798E-10
	ko00830	Retinol metabolism	1.02021E-12	3.43811E-10
	ko00140	Steroid hormone biosynthesis	1.78094E-12	6.00176E-10
	ko00980	Metabolism of xenobiotics by cytochrome P450	1.366E-11	4.60342E-09
	ko05204	Chemical carcinogenesis	2.1777E-11	7.33885E-09
	ko05332	Graft-versus-host disease	4.49901E-11	1.51617E-08
	ko05330	Allograft rejection	2.90528E-10	9.7908E-08
	ko05322	Systemic lupus erythematosus	2.34432E-09	7.90037E-07
	ko04630	Jak-STAT signaling pathway	3.79541E-09	1.27905E-06
	ko04940	Type I diabetes mellitus	4.35455E-08	1.46748E-05
	ko04612	Antigen processing and presentation	6.92961E-08	2.33528E-05
	ko00983	Drug metabolism - other enzymes	2.02365E-07	6.81971E-05
	ko05167	Kaposi sarcoma-associated herpesvirus infection	3.04773E-07	0.000102708
	ko04915	Estrogen signaling pathway	9.41846E-07	0.000317402
	ko05169	Epstein-Barr virus infection	1.35327E-06	0.000456051
	ko05168	Herpes simplex infection	2.27644E-06	0.000767161
	ko04623	Cytosolic DNA-sensing pathway	5.41762E-06	0.001825739
	ko05170	Human immunodeficiency virus 1 infection	1.23868E-05	0.004174343
	ko05034	Alcoholism	1.54525E-05	0.005207497
	ko04060	Cytokine-cytokine receptor interaction	1.65324E-05	0.005571405
	ko05163	Human cytomegalovirus infection	2.56805E-05	0.008654339
	ko05202	Transcriptional misregulation in cancer	6.9027E-05	0.023262095
	ko05416	Viral myocarditis	0.000107718	0.036300843
Pinnipedia	ko04740	Olfactory transduction	2.39611E-88	8.07488E-86
	ko04622	RIG-I-like receptor signaling pathway	1.24333E-08	4.19001E-06
	ko05320	Autoimmune thyroid disease	6.25557E-07	0.000210813
	ko00053	Ascorbate and aldarate metabolism	1.34674E-06	0.000453853
	ko00040	Pentose and glucuronate interconversions	1.30995E-05	0.004414527
	ko04623	Cytosolic DNA-sensing pathway	4.17329E-05	0.014063972
Manatee	ko04740	Olfactory transduction	3.9273E-163	3.9273E-163

Table S17. KEGG enrichment of significantly expanded gene families in marine mammals.

Groups	ID	Description	p-value	Adjusted p-value
Cetacean	ko01503	Cationic antimicrobial peptide (CAMP) resistance	2.01794E-84	6.80047E-82
	ko03010	Ribosome	3.5973E-63	1.21229E-60
	ko03040	Spliceosome	5.60995E-16	1.89055E-13
	ko04217	Necroptosis	4.56869E-14	1.53965E-11
	ko00260	Glycine, serine and threonine metabolism	2.27241E-13	7.65801E-11
	ko00630	Glyoxylate and dicarboxylate metabolism	3.90817E-13	1.31705E-10
	ko00563	Glycosylphosphatidylinositol (GPI)-anchor biosynthesis	1.11349E-11	3.75247E-09
	ko04260	Cardiac muscle contraction	1.42996E-11	4.81896E-09
	ko05012	Parkinson disease	1.32807E-10	4.47558E-08
	ko04714	Thermogenesis	2.09499E-09	7.06012E-07
	ko00190	Oxidative phosphorylation	3.90343E-09	1.31546E-06
	ko04932	Non-alcoholic fatty liver disease (NAFLD)	1.41345E-08	4.76334E-06
	ko05016	Huntington disease	3.53843E-07	0.000119245
	ko05010	Alzheimer disease	1.54867E-06	0.000521903
Pinnipedia	ko00640	Propanoate metabolism	1.26892E-89	4.27627E-87
	ko04626	Plant-pathogen interaction	6.24118E-85	2.10328E-82
	ko04922	Glucagon signaling pathway	8.34171E-85	2.81115E-82
	ko00620	Pyruvate metabolism	1.31689E-64	4.43794E-62
	ko00270	Cysteine and methionine metabolism	1.241E-62	4.18218E-60
	ko04016	MAPK signaling pathway - plant	5.93664E-53	2.00065E-50
	ko00010	Glycolysis / Gluconeogenesis	1.56986E-44	5.29043E-42
	ko05418	Fluid shear stress and atherosclerosis	3.52963E-41	1.18948E-38
	ko04745	Phototransduction - fly	8.11623E-37	2.73517E-34
	ko05010	Alzheimer disease	7.98437E-35	2.69073E-32
	ko04744	Phototransduction	5.91371E-34	1.99292E-31
	ko00190	Oxidative phosphorylation	2.4849E-27	8.37413E-25
	ko04924	Renin secretion	8.26111E-19	2.78399E-16
	ko05031	Amphetamine addiction	5.59388E-18	1.88514E-15
	ko05133	Pertussis	5.59388E-18	1.88514E-15
	ko04720	Long-term potentiation	1.83914E-17	6.19791E-15
	ko05214	Glioma	3.25742E-17	1.09775E-14
	ko03013	RNA transport	3.62975E-17	1.22323E-14
	ko04970	Salivary secretion	2.77901E-16	9.36527E-14
	ko04971	Gastric acid secretion	4.59501E-16	1.54852E-13
	ko03040	Spliceosome	3.47636E-15	1.17153E-12
	ko04915	Estrogen signaling pathway	1.49966E-14	5.05387E-12
	ko04912	GnRH signaling pathway	3.82254E-14	1.28819E-11
	ko05012	Parkinson disease	7.56162E-14	2.54827E-11
	ko04925	Aldosterone synthesis and secretion	4.93446E-13	1.66291E-10
	ko04713	Circadian entrainment	9.59522E-13	3.23359E-10
	ko04750	Inflammatory mediator regulation of TRP	9.59522E-13	3.23359E-10

		channels		
	ko04625	C-type lectin receptor signaling pathway	1.32486E-12	4.46477E-10
	ko04714	Thermogenesis	1.71327E-12	5.77373E-10
	ko04070	Phosphatidylinositol signaling system	1.81775E-12	6.12582E-10
	ko04916	Melanogenesis	6.068E-12	2.04491E-09
	ko04114	Oocyte meiosis	1.38128E-10	4.65492E-08
	ko04722	Neurotrophin signaling pathway	2.18794E-10	7.37337E-08
	ko04371	Apelin signaling pathway	1.19242E-09	4.01847E-07
	ko04270	Vascular smooth muscle contraction	1.45223E-09	4.894E-07
	ko04910	Insulin signaling pathway	1.45223E-09	4.894E-07
	ko05016	Huntington disease	2.69419E-09	9.07942E-07
	ko04261	Adrenergic signaling in cardiomyocytes	5.31928E-09	1.7926E-06
	ko05134	Legionellosis	5.6881E-09	1.91689E-06
	ko04728	Dopaminergic synapse	1.0534E-08	3.54997E-06
	ko04921	Oxytocin signaling pathway	2.34619E-08	7.90668E-06
	ko04218	Cellular senescence	4.95219E-08	1.66889E-05
	ko05152	Tuberculosis	9.95737E-08	3.35563E-05
	ko05167	Kaposi sarcoma-associated herpesvirus infection	1.13884E-07	3.83788E-05
	ko04022	cGMP-PKG signaling pathway	2.78271E-07	9.37774E-05
	ko04740	Olfactory transduction	1.92891E-06	0.000650044
	ko05170	Human immunodeficiency virus 1 infection	5.11941E-06	0.001725242
	ko04024	cAMP signaling pathway	8.56718E-06	0.002887138
	ko04020	Calcium signaling pathway	1.09529E-05	0.003691124
	ko04015	Rap1 signaling pathway	1.18678E-05	0.003999463
	ko05034	Alcoholism	1.50249E-05	0.005063406
	ko05163	Human cytomegalovirus infection	1.50249E-05	0.005063406
	ko04014	Ras signaling pathway	4.73211E-05	0.015947215
Manatee	ko05322	Systemic lupus erythematosus	3.76568E-90	1.26903E-87
	ko04612	Antigen processing and presentation	5.95247E-86	2.00598E-83
	ko05034	Alcoholism	1.26458E-51	4.26164E-49
	ko05320	Autoimmune thyroid disease	9.47979E-42	3.19469E-39
	ko04650	Natural killer cell mediated cytotoxicity	7.18364E-39	2.42089E-36
	ko05332	Graft-versus-host disease	3.96245E-27	1.33535E-24
	ko05330	Allograft rejection	1.21595E-23	4.09777E-21
	ko04217	Necroptosis	4.52294E-22	1.52423E-19
	ko04940	Type I diabetes mellitus	6.79793E-22	2.2909E-19
	ko05416	Viral myocarditis	7.7342E-22	2.60643E-19
	ko05168	Herpes simplex infection	6.4271E-18	2.16593E-15
	ko04145	Phagosome	5.74367E-17	1.93562E-14
	ko05164	Influenza A	3.57063E-16	1.2033E-13
	ko05203	Viral carcinogenesis	2.93622E-15	9.89507E-13
	ko04915	Estrogen signaling pathway	6.35424E-11	2.14138E-08
	ko05169	Epstein-Barr virus infection	1.5562E-10	5.2444E-08
	ko05310	Asthma	5.46246E-09	1.84085E-06
	ko05150	Staphylococcus aureus infection	1.66896E-08	5.62439E-06
	ko05162	Measles	1.01975E-07	3.43654E-05
	ko05145	Toxoplasmosis	2.17715E-07	7.33701E-05

	ko04514	Cell adhesion molecules (CAMs)	8.50048E-07	0.000286466
	ko04621	NOD-like receptor signaling pathway	2.25457E-06	0.000759789
	ko05323	Rheumatoid arthritis	4.31049E-06	0.001452636
	ko05167	Kaposi sarcoma-associated herpesvirus infection	5.40767E-06	0.001822383
	ko00980	Metabolism of xenobiotics by cytochrome P450	9.33341E-06	0.003145359
	ko00982	Drug metabolism - cytochrome P450	1.34635E-05	0.004537199
	ko04622	RIG-I-like receptor signaling pathway	2.00725E-05	0.006764426
	ko04672	Intestinal immune network for IgA production	3.67469E-05	0.012383702
	ko05152	Tuberculosis	8.01766E-05	0.0270195
	ko05163	Human cytomegalovirus infection	9.66036E-05	0.032555416
	ko04623	Cytosolic DNA-sensing pathway	0.000116763	0.039349071

Table S18. The genes under positive selection on the ancestral branches of three marine mammal lineages.

Group	Cluster	Gene	p-value	Adjusted p-value
Cetacean	XP_013829733	*SMCHD1*	0	0
	XP_005690745	*TARDBP*	0.000000001	6.0875E-06
	XP_017920316	*GPR155*	0.000000194	0.000787317
	XP_017909135	*DPP8*	0.000000544	0.0016558
	XP_017904280	*MGAT3*	0.000007031	0.017120485
Pinniped	XP_017908150	*COL27A1*	0.000000001	0.000009204
	XP_017900031	*MROH7*	0.000000004	0.000018408
	XP_017902954	*CRYGN*	0.000000029	0.000088972
	XP_017914185	*PLEC*	0.000000067	0.000154167
	XP_005689035	*RP1*	0.000001519	0.002796175
	XP_017914796	*MYO7A*	0.000002944	0.004516096
	XP_017910194	*IL18RAP*	0.000007232	0.009509047
	XP_017913603	*PHC2*	0.000009682	0.011139141
	XP_013826551	*LOC102177559*	0.000017297	0.017689065
	XP_017909486	*PAPLN*	0.000032228	0.029662651
	XP_017905078	*USO1*	0.000041911	0.035068077
Manatee	XP_017900554	*HCFC1*	0	0
	XP_017908431	*SOGA3*	0	0
	XP_017914185	*PLEC*	0	0
	XP_017921499	*DYNC1H1*	0	0
	XP_005691237	*MAB21L2*	0.000000067	0.000116098
	XP_013817743	*TMEM121*	0.000000301	0.000434644
	XP_017897223	*HMX3*	0.000002383	0.002949473
	XP_017921784	*EFL1*	0.000003153	0.003414699
	XP_017899314	*MED14*	0.000007601	0.007317229
	XP_017894857	*PHLPP1*	0.000010184	0.008823418
	XP_005688575	*MYL9*	0.000012671	0.00998014
	XP_017899469	*PHKA2*	0.000018262	0.013185164
	XP_017914562	*SSRP1*	0.000026486	0.016595892
	XP_017918057	*RELB*	0.000026817	0.016595892
	XP_017896249	*LOC102177581*	0.000033085	0.019109896
	XP_005697770	*RNF40*	0.000045784	0.024792036

Table S20. KEGG/GO enrichment of rapidly evolving genes in the West Indian manatee.

KO	Info	p-value	Adjusted p-value
ko04971	Gastric acid secretion	1.25E-07	4.22834E-05
GO	**Function**	**p-value**	**Adjusted p-value**
GO:0015267	channel activity	2.96E-14	1.80771E-10
GO:0022803	passive transmembrane transporter activity	2.96E-14	1.80771E-10
GO:0005216	ion channel activity	6.99E-14	4.27737E-10
GO:0022838	substrate-specific channel activity	6.99E-14	4.27737E-10
GO:0005215	transporter activity	1.26E-13	7.68284E-10
GO:0022857	transmembrane transporter activity	3.67E-13	2.24457E-09
GO:0016594	glycine binding	6.75E-13	4.12723E-09
GO:0006811	ion transport	2.41E-12	1.47195E-08
GO:0022892	substrate-specific transporter activity	6.36E-12	3.89021E-08
GO:0022891	substrate-specific transmembrane transporter activity	1.22E-11	7.43502E-08
GO:0055085	transmembrane transport	3.39E-11	2.07481E-07
GO:0015075	ion transmembrane transporter activity	4.12E-11	2.52202E-07
GO:0006810	transport	1.31E-10	8.01257E-07
GO:0051234	establishment of localization	1.41E-10	8.62656E-07
GO:0022836	gated channel activity	3.22E-10	1.97056E-06
GO:0015278	calcium-release channel activity	3.85E-09	2.35736E-05
GO:0015276	ligand-gated ion channel activity	4.85E-09	2.96893E-05
GO:0022834	ligand-gated channel activity	4.85E-09	2.96893E-05
GO:0003779	actin binding	2.01E-08	0.000123152
GO:0006812	cation transport	2.11E-08	0.000129204
GO:0005261	cation channel activity	2.66E-08	0.00016282
GO:0030001	metal ion transport	3.97E-08	0.000242627
GO:0046873	metal ion transmembrane transporter activity	1.86E-07	0.00114006
GO:0005253	anion channel activity	3.1E-07	0.001894354
GO:0015108	chloride transmembrane transporter activity	3.1E-07	0.001894354
GO:0005217	intracellular ligand-gated ion channel activity	4.18E-07	0.002559453
GO:0005219	ryanodine-sensitive calcium-release channel activity	4.32E-07	0.002640313
GO:0015103	inorganic anion transmembrane transporter activity	6.89E-07	0.004216182
GO:0008092	cytoskeletal protein binding	7.52E-07	0.004600058
GO:0022890	inorganic cation transmembrane transporter activity	1.14E-06	0.006958654
GO:0034220	ion transmembrane transport	1.28E-06	0.007841804
GO:0043244	regulation of protein complex disassembly	1.46E-06	0.008954673
GO:0008324	cation transmembrane transporter activity	1.56E-06	0.0095629
GO:0009987	cellular process	2.77E-06	0.016956957
GO:0015672	monovalent inorganic cation transport	3.24E-06	0.019803139
GO:0008509	anion transmembrane transporter activity	3.26E-06	0.019930922
GO:0005254	chloride channel activity	4.37E-06	0.026738977
GO:0005237	inhibitory extracellular ligand-gated ion channel activity	5.07E-06	0.030984161
GO:0008091	spectrin	5.07E-06	0.030984161
GO:0016933	extracellular-glycine-gated ion channel activity	5.07E-06	0.030984161
GO:0016934	extracellular-glycine-gated chloride channel activity	5.07E-06	0.030984161

Table S25. KEGG enrichment of convergent genes in Cetaceans/Sirenia.

Term	p-value	Genes	Adjusted p-value
chx04728:Dopaminergic synapse	0.0041	*KCNJ5, GRIN2A, KIF5B, PPP2R3A, MAPK13*	0.51
chx04015:Rap1 signaling pathway	0.0120	*MAGI1, GRIN2A, SIPA1L1, THBS1, MAPK13*	1.00
chx04611:Platelet activation	0.0270	*COL11A1, COL5A2, PLA2G4A, MAPK13*	1.00

Table S26. *UCP1* status in marine mammals.

Order	Family	Latin name	Common name	Status	Source
Carnivora	*Phocidae*	*Neomonachus schauinslandi*	Hawaiian monk seal	Normal	NCBI
		Pusa sibirica	Baikal seal	Normal	New sequencing
		Phoca largha	Spotted seal	Normal	New sequencing
	Otariidae	*Callorhinus ursinus*	Northern fur seal	Normal	NCBI
		Arctocephalus gazella	Antarctic fur seal	Pseudogene	NCBI
		Arctocephalus australis	South American fur seal	Normal	New sequencing
		Zalophus californianus	California sea lion	Normal	New sequencing
		Otaria byronia	South American sea lion	Normal	New sequencing
Artiodactyla	*Balaenopteridae*	*Balaenoptera brydei*	Bryde's whale	Pseudogene	New sequencing
		Balaenoptera musculus	Blue whale	Pseudogene	New sequencing
	Eschrichtiidae	*Eschrichtius robustus*	Gray whale	Pseudogene	NCBI
	Kogiidae	*Kogia breviceps*	Pygmy sperm whale	Pseudogene	New sequencing
		Kogia sima	Dwarf sperm whale	Pseudogene	New sequencing
	Ziphiidae	*Mesoplodon densirostris*	Blainville's beaked whale	Lost	New sequencing
	Phocoenidae	*Phocoena phocoena*	Harbour porpoise	Partially lost	NCBI
		Neophocaena asiaeorientalis	Narrow-ridged finless porpoise	Partially lost	NCBI
	Delphinidae	*Lagenorhynchus australis*	Peale's dolphin	Lost	New sequencing
		Lagenorhynchus obliquidens	Pacific white-sided dolphin	Lost	New sequencing
		Grampus griseus	Risso's dolphin	Lost	New sequencing
		Pseudorca crassidens	False killer whale	Lost	New sequencing
		Tursiops aduncus	Indo-Pacific bottlenose dolphin	Lost	New sequencing
		Sousa chinensis	Indo-Pacific humpback dolphin	Lost	CNGB

Table S28. Specific amino acid mutations in the deep-diving species.

Gene	Variation Numbers	Amino Acid Sites
SETX	4	D2324E, D1979Y, A1672P, K1057N
NOS2	1	T656A
ABL1	1	S1127G
ABLIM1	1	A200T
ADD1	1	K369R
CDA	1	C22S
CEP170	1	A413T
COL2A1	1	A500V
CP	1	R781Q
CYB5A	1	L37V
DHCR7	1	T69M
DIDO1	1	I1207T
DSP	1	V1185A
F11R	1	Q214R
GBE1	1	L227V
GPR146	1	V201A
GPRIN3	1	G622S
HYOU1	1	R193H
IL6ST	1	K206N
ITPR2	1	T2478M
LEPR	1	S389A
LIMCH1	1	G352S
MB	1	N13H
ME2	1	G178A
MPHOSPH9	1	T1142S
NR3C2	1	N43K
OBSCN	1	E2816Q
PFN1	1	S133C
PKD2	1	I435V
PLD1	1	V618I
RAB3GAP2	1	A1174T
RERE	1	K48R
SDPR	1	S108G
SNRNP40	1	G9A
SRPRB	1	G13D
TCF25	1	E525K
TMEM9	1	I94V
TNFRSF11B	1	R138Q
TRPV3	1	M19V

Table S29. The PSGs and REGs in the deep-diving species.

Type	Cluster	Gene	p-value	Adjusted p-value
PSGs	XP_017911524	*SETX*	1.084E-06	4.18×10^{-4}
	XP_013820122	*TMPRSS11D*	9.211E-05	6.46×10^{-3}
	XP_005690391	*GIF*	0.0004856	1.97×10^{-2}
REGs	XP_005695214	*STOML1*	0	0
	XP_017894273	*DSP*	4E-09	3.36×10^{-6}
	XP_005690437	*ZC3H11A*	3.6E-08	2.02×10^{-5}
	XP_017911524	*SETX*	6.5E-08	2.73×10^{-5}
	XP_017906450	*HNRNPM*	2.12E-07	7.12×10^{-5}
	XP_017899142	*UBA1*	1.043E-06	2.92×10^{-4}
	XP_017894157	*DST*	5.058E-06	1.21×10^{-3}
	XP_005684064	*LOXL2*	1.065E-05	2.24×10^{-3}
	XP_013820122	*TMPRSS11D*	0.00013	2.64×10^{-3}
	XP_017897672	*ANK3*	2.746E-05	5.13×10^{-3}
	XP_017903249	*PLD1*	3.418E-05	5.74×10^{-3}
	XP_017915751	*CEP170*	8.861E-05	1.21×10^{-2}
	XP_005681363	*ENPEP*	8.908E-05	1.21×10^{-2}
	XP_017920162	*ITGA6*	9.396E-05	1.21×10^{-2}
	XP_013829007	*ITPR1*	0.0001192	1.43×10^{-2}
	XP_017899149	*USP11*	0.0001349	1.44×10^{-2}
	XP_017919249	*EXOC7*	0.0001375	1.44×10^{-2}
	XP_013831518	*DHCR7*	0.0001765	1.69×10^{-2}
	XP_017915483	*PPFIA4*	0.0001808	1.69×10^{-2}
	XP_005699122	*RHOBTB1*	0.0002008	1.69×10^{-2}
	XP_017913154	*RBCK1*	0.0002171	1.69×10^{-2}
	XP_017908858	*SYNE1*	0.0002191	1.69×10^{-2}
	XP_005676318	*TMEM185B*	0.0002216	1.69×10^{-2}
	XP_005690391	*GIF*	0.0015898	1.69×10^{-2}
	XP_017894508	*BAG2*	0.0002432	1.77×10^{-2}
	XP_005675342	*ACTL6A*	0.0002522	1.77×10^{-2}
	XP_017895166	*EPB41L3*	0.0003213	2.10×10^{-2}
	XP_017916569	*SART3*	0.0003245	2.10×10^{-2}
	NP_001272656	*TYRP1*	0.0004935	3.07×10^{-2}
	XP_017897599	*MYOM2*	0.0005688	3.41×10^{-2}
	XP_017898587	*CFL1*	0.0005961	3.45×10^{-2}
	XP_005696106	*LARS2*	0.0006708	3.59×10^{-2}
	XP_017894721	*TPMT*	0.0006791	3.59×10^{-2}
	XP_017894949	*DSC3*	0.0006835	3.59×10^{-2}
	XP_005701359	*ALAS2*	0.0007208	3.60×10^{-2}
	XP_017908923	*IGF2R*	0.0007292	3.60×10^{-2}
	XP_005676103	*DPP4*	0.0008152	3.91×10^{-2}
	XP_013831900	*PQBP1*	0.0008423	3.93×10^{-2}
	XP_017909607	*GBE1*	0.000887	4.03×10^{-2}
	XP_005694116	*P4HB*	0.0009856	4.36×10^{-2}
	XP_017908696	*EPB41L2*	0.0010847	4.59×10^{-2}
	XP_017916467	*GAL3ST1*	0.0010938	4.59×10^{-2}
	XP_005690423	*ADORA1*	0.0011303	4.59×10^{-2}
	XP_017917832	*RYR1*	0.0011464	4.59×10^{-2}

Table S31. KEGG enrichment of rapidly evolving genes in LSWG.

KO	Pathway	p-value	Adjusted p-value
ko04340	Hedgehog signaling pathway	5.14×10^{-9}	1.73×10^{-6}
ko04310	Wnt signaling pathway	1.64×10^{-6}	5.52×10^{-4}
ko05205	Proteoglycans in cancer	1.43×10^{-4}	4.83×10^{-2}
ko04512	ECM-receptor interaction	3.98×10^{-5}	1.34×10^{-2}

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